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UNDERWATER SPECIES CLASSIFICATION USING CNN: ACASE STUDY ON SEA ANIMALS

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CERTIFICATE

This is to certify that the Major Project report entitled "**UNDERWATER SPECIES CLASSIFICATION USING CNN: A CASE STUDY ON SEA ANIMALS**" is the Bonafide work carried out and submitted by

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To the department of Computer Science and Engineering, **Malla Reddy Engineering College and Management Sciences**, in partial fulfilment for the award of **BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING (AI & ML)** during the academic year 2023-2024.

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.

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ABSTRACT

Underwater species classification is a critical task in marine biology and ecological conservation efforts. The underwater environment poses unique difficulties such as varying lighting conditions, occlusions, and diverse species morphology, making traditional classification methods inefficient. This study focuses on ten commonly encountered marine species, namely Fish, Goldfish, Harbor seal, Jellyfish, Lobster, Oyster, Sea turtle, Squid, Starfish, and others. The conventional systems employed for underwater species classification often rely on handcrafted features and shallow machine learning models, leading to limited accuracy and generalization in complex underwater scenarios. These systems struggle to adapt to variations in species appearance and environmental factors, hampering their practicality for real-world applications. Moreover, manual feature engineering is labor- intensive and lacks scalability. In contrast, the proposed system leverages the power of CNNs, a class of deep learning models specifically designed for image analysis tasks. CNNs have demonstrated remarkable success in image recognition tasks due to their ability to automatically learn hierarchical features from raw pixel data. Our system employs a state-of-the-art CNN architecture, pre-trained on large-scale underwater image datasets, to extract meaningful features from marine organism images. Transfer learning techniques are utilized to fine-tune the CNN on the specific task of marine species classification. The proposed system addresses the drawbacks of conventional approaches by offering improved accuracy, robustness to environmental variations, and reduced reliance on manual feature engineering.

Keywords : Underwater species, Convolutional Neural Network (CNN), Machine learning models, Transfer learning technique.

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LIST OF ABREVATIONS

CNN - CONVOLUTIONAL NEURAL NETWORKS

ROV - REMOTELY OPERATED VEHICLES

AUV - AUTONOMOUS UNDERWATER VEHICLES

AR - AUGUMENTED REALITY

VR - VIRTUAL REALITY

TP - TRUE POSITIVES

TN - TRUE NEGATIVES

FP - FALSE POSITIVES

FN - FALSE NEGATIVES

DL - DEEP LEARNING

ML - MACHINE LEARNING

SSIM - STRUCTURAL SIMILARITY INDEX

PSNR - PEAK SIGNAL TO NOISE RATIO

UIQM - UNDERWATER IMAGE QUALITY MEASURES

UCIQE - UNDERWATER COLOUR IMAGE QUALITY EVALUATION

YOLO - YOU ONLY LOOK ONCE

BRUVS - BAITED REMOTE UNDERWATER VEDIO STATION

DNN - DEEP NEURAL NETWORK

FSC - FISH SPECIES CLASSIFICATION

GRAD-CAM - GRADIENT-WEIGHTED CATEGORY ACTIVATION MAPPING

FPN - FEATURE PYRAMID NETWORK

I-PANet - IMPROVED PATH AGGREGATION NETWORK

EGWO - ENHANCED GREY WOLF OPTIMIZATION

SACovamnet - SANDWICH ATTENTION CONVARIANCE MATRIC NETWORK

GAP - GLOBAL AVERAGE POOLING

FL - FOCAL LOSS

SE - SQUEEZE - AND - EXCITATION

ViT - VISION TRANSFORMER

TCPFL - TWISTED AND COILED POLYMER FISHING LINE MUSCLE

SNR - SIGNAL-TO-NOISE RATIO

UUV - UNMANNED UNDERWATER VEHICLES

PCM - PHASE CHANGE MATERIALS

NARW - NORTH ATLANTIC RIGHT WHALE

AIMS - ACOUSTIC IMAGING MICROTOME SYSTEM

FFM - FEATURE FUSION MODULE

HABs - HIGH-ALTITUDE BIOLOGICAL SPECIES

UFO - UNDERWATER FISH OBSERVATORY

CADDY - COGNITIVE AUTONOMOUS DIVING BUDDY

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Chapter 1

Introduction

1.1 History

The history of classifying underwater species spans millennia, reflecting humanity's fascination with the ocean's mysteries. The background of identifying marine varieties is an intricate procedure affected by the growth of aquatic biology as a clinical self- control, technical developments plus the developing understanding of aquatic communities as shown in Figure 1. Very early monitoring and also categories of sea life were made by old seafarers along with botanists such as Aristotle. European travellers like James Cook as well as Charles Darwin accumulated and also catalogued marine samplings throughout the 15th-17th century which resulted in the growth of Darwin's concept of development.

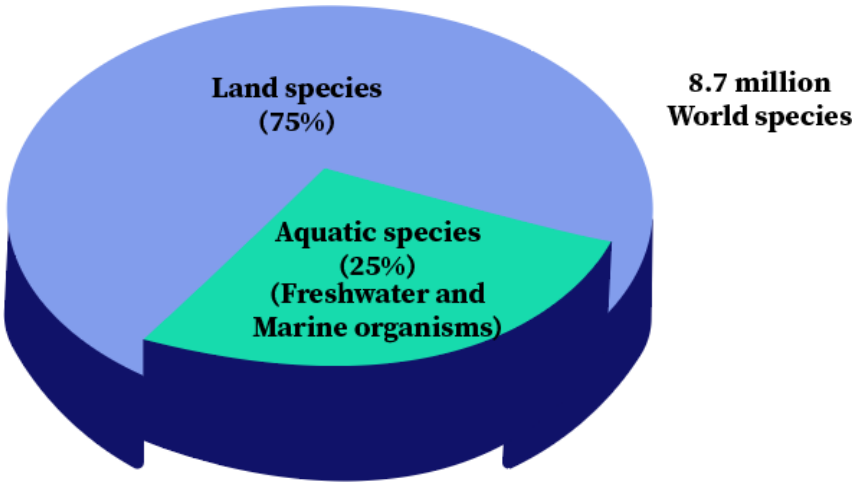


Figure 1.1 Sample Underwater Species Statistics.

The 19th century saw substantial progression in aquatic expedition along with category with the development of the fish tank in the 1850s and also the Challenger exploration noting the initial significant initiative to examine sea biology and also location methodically. The 20th century brought extensive adjustments with the arrival of scuba diving, from another location automobiles.

1.2 Motivation

Underwater species classification is crucial for various reasons, including conservation, biodiversity, ecosystem health monitoring, fisheries management, invasive species control, marine research, climate change studies, pollution impact assessment, public education, economic value, and cultural and recreational importance. Accurate classification helps in assessing pollution levels, developing remediation strategies, and enhancing cultural and recreational activities like diving and snorkelling. Marine species can also act as indicators of climate change, reflecting changes in ocean temperatures and acidity.

Underwater species classification is essential for maintaining healthy marine ecosystems, supporting sustainable use of marine resources, and furthering our scientific understanding of the oceans. It helps in understanding the roles of marine life, their conservation status, and the need for protection measures, as well as monitoring the health of marine ecosystems and assessing pollution impacts.

Underwater species classification is a complex process that involves scientific discovery, understanding, conservation, sustainable resource management, environmental health monitoring, medical and biotechnological research, public awareness, global climate regulation, and cultural and aesthetic value. The ocean, covering over 70% of Earth's surface, is home to a vast array of life forms, each providing unique insights into evolutionary biology, ecological interactions, and Earth's history. Studying these spcies helps in sustainable management, ensuring resources are not overexploited and marine ecosystems remain healthy and productive. Underwater species classification raises public awareness about the richness and importance of marine life, providing educational opportunities, and inspiring future generations to appreciate and protect the oceans.

1.3 Traditional systems

The traditional system for underwater species classification in marine ecosystem is classified through manually by well-trained experts and high labour. Human Vision System is one of the methods, in this the classification is based on eye vision experience. It is based on both eye and brain. But only limited species was classified and for the classified species record with be taken manually. Domain experts-based classification is one of the manual methods in which the classifying or categorizing data, objects, or phenomena based on the knowledge and

experience of experts in a specific field or domain. Experts use their knowledge to manually identify and classify species. This might involve observing physical characteristics, behaviours, or other specific traits that are known to differentiate species.

Early naturalists and marine biologists relied on direct observation of marine life, often from shorelines, boats, or shallow waters. They observed physical characteristics, behaviours, and habitats of marine species. Specimen collection was common, with specimens preserved and brought back for further study. Morphological analysis was the primary method for classifying species, examining the physical form and structure of an organism. Dissection and microscopic examination were used to study the internal anatomy of larger marine organisms, providing insights into their biology and taxonomy. Accurate illustration and detailed descriptive writing were crucial for documenting species, with detailed drawings and descriptions shared with the scientific community.

The development of basic diving equipment, like diving bells and suits, allowed for more direct exploration of underwater environments. Early classification systems were based on visible traits and similarities among organisms, with Carl Linnaeus' binomial nomenclature system being the most influential. Collaboration and exchange of specimens were crucial for building comprehensive collections and expanding knowledge about marine biodiversity. Although these traditional methods had limitations in depth accessibility, species identification accuracy, and ecological understanding, they were instrumental in the initial exploration and documentation of marine life.

1.4 Problem statement

Traditional methods of underwater species classification face numerous challenges, including limited speed and efficiency, high labour and expert requirements, inconsistency, scalability issues, limited depth and breadth of analysis, difficulty in challenging conditions, cumbersome data recording and management processes, limited real-time analysis capability, restricted access to remote or hazardous locations, and potential impact on marine life. These issues highlight the need for more advanced, efficient, and reliable methods, such as the use of AI and machine learning techniques like Convolutional Neural Networks (CNNs).

Traditional methods are time-consuming and rely on manual observation and analysis, making them time-consuming and inefficient. Accurate species

identification requires highly trained experts, which was costly and may not always be readily available, especially in remote or less-studied regions. Manual classification is prone to human error and can vary between observers, leading to inconsistencies in data.

Traditional methods are not easily scalable to large-scale studies or continuous monitoring due to the labour- intensive nature of the work. Human observers can only process a limited amount of information at a time, which may lead to missed details and inconsistencies in data. Direct human interactions such as diving or capturing specimens for identification can potentially disturb marine habitats and organisms.

To overcome these challenges, the CNN method is proposed for image classification, which is inexpensive, time-consuming, and has no limit for species classification. Marine biodiversity is a complex issue that requires a multidisciplinary approach, including scientific research, technological innovation, policy formulation, and public engagement. Accurate identification of marine species, particularly those that are morphologically similar or cryptic, is crucial for assessing and predicting their impacts on marine biodiversity and ecosystem health.

1.5 Research objective

- The main objective of this project is classifying the species in water. There are different kinds of species in different diversities.
- This classification is very useful for ecological studies, environmental monitoring, marine resource management, fisheries, etc.
- This research can lead to a better understanding of ecosystem dynamics, species relationships, and environmental impacts on marine life.
- The development and refinement of techniques for underwater species classification, particularly using machine learning and AI, contribute to advancements in marine science technology. This can lead to more efficient and accurate research methods.

1.6 Applications

Automated underwater surveys: CNNs can analyse video footage from underwater drones or ROVs to identify and count different fish species, coral types, and other marine organisms in real-time. This data is crucial for monitoring ecosystem health, tracking biodiversity, and detecting invasive species.

Species identification and protection: By classifying underwater features like kelp forests, seagrass meadows, and coral reefs, CNNs can help map sensitive marine habitats and inform conservation efforts. Real-time monitoring can also detect threats like pollution or illegal fishing activities.

Fish stock assessment and management: CNNs was used to estimate fish populations in real-time, aiding in sustainable fisheries management. This can help prevent overfishing and ensure healthy fish stocks.

Disease detection in species: CNNs can analyse fish images to identify diseases or parasites, allowing for early intervention and improved fish health in aquaculture farms.

Remote Sensors and Satellite Imagery: Satellites equipped with advanced sensors can capture large-scale images of the ocean surface. These images was analysed to identify phytoplankton blooms, which are indicative of certain underwater species, or to study coral reef systems.

Underwater Drones and ROVs: Remotely Operated Vehicles (ROVs) and autonomous underwater vehicles (AUVs) equipped with cameras and sensors can gather data and images from underwater environments. The collected data was used to classify and monitor underwater species in their natural habitats.

Environmental DNA Analysis: This method involves analysing water samples for DNA fragments shed by organisms. Bioinformatics tools are used to identify and

classify species based on these DNA fragments. This is particularly useful for detecting rare or elusive species.

Sound Monitoring: Underwater microphones (hydrophones) can capture the sounds of marine life. These sounds was analysed using signal processing techniques to identify and classify species, which is especially useful for marine mammals like whales and dolphins.

Educational and Training Tools: Augmented reality (AR) and virtual reality (VR) applications can simulate underwater environments for educational purposes, helping students and researchers learn about species identification and classification in an interactive way.

Genetic Analysis: Advanced genomic sequencing technologies enable detailed genetic analysis of species. Bioinformatics tools are used to classify species based on their genetic makeup, which is especially useful for studying microorganisms and cryptic species.

1.7 Advantages

Automated and Fast Processing: Manual classification of underwater species is time consuming and requires expert knowledge. CNNs can automate this process by analysing large volumes of data much faster than humans can.

Reduced Bias: Manual classification was subjective and influenced by the observer's experience and fatigue. CNNs offer a more objective approach.

Feature Learning: CNNs automatically learn and identify features relevant for classification. This is especially useful in underwater environments where certain species characteristics might not be immediately obvious to human observers.

High Accuracy: CNNs are capable of achieving high accuracy in image classification tasks due to their ability to learn hierarchical feature representations. This is crucial in underwater environments where species may have subtle differences.

Improved Understanding of Marine Biodiversity: Efficient and accurate classification can lead to better understanding of marine ecosystems, aiding in conservation efforts and the study of marine biodiversity.

Access to Remote and Deep-Sea Areas: Underwater drones and ROVs can explore areas that are otherwise inaccessible to humans, such as deep-sea ecosystems, increasing our knowledge of these largely unexplored areas.

Enhanced Biodiversity Monitoring: These technologies enable more comprehensive biodiversity assessments, helping in the detection of invasive species, monitoring of endangered species, and understanding of ecosystem dynamics.

Long-Term Monitoring and Trend Analysis: Continuous monitoring using these technologies allows for the tracking of long-term trends in marine ecosystems, essential for understanding the impacts of environmental changes and human activities.

Cost-Effectiveness: Although the initial investment in technology was high, in the long run, these methods was more cost-effective than traditional techniques, especially considering the scale and depth of data they provide.

1.8 Performance metrics

Confusion Matrix: It is the easiest way to measure the performance of a classification problem where the output was of two or more type of classes. A confusion matrix is nothing but a table with two dimensions viz. "Actual" and "Predicted" and furthermore, both the dimensions have "True Positives (TP)", "True Negatives (TN)", "False Positives (FP)", "False Negatives (FN)" as shown below −

		Actual	
		1	0
Predicted	1	True Positives (TP)	False Positives (FP)
	0		True Negatives (TN)

Figure 1.8. Confusion matrix.

- **True Positives (TP)** − It is the case when both actual class & predicted class of data point is 1.
- **True Negatives (TN)** − It is the case when both actual class & predicted class of data point is 0.
- **False Positives (FP)** − It is the case when actual class of data point is 0 & predicted class of data point is 1.
- **False Negatives (FN)** − It is the case when actual class of data point is 1 & predicted class of data point is 0.

Classification report: : The classification report visualizer displays the precision, recall, F1, and support scores for the model. In order to support easier interpretation and problem detection, the report integrates numerical scores with a color-coded heatmap. All heatmaps are in the range (0.0, 1.0) to facilitate easy comparison of classification models across different classification reports.

Accuracy: It is most common performance metric for classification algorithms. It was defined as the number of correct predictions made as a ratio of all predictions made.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision: Precision, used in document retrievals, was defined as the number of correct documents returned by our ML model.

$$Precision = \frac{TP}{TP+FP}$$

Recall: Recall was defined as the number of positives returned by our ML model.

$$Recall = \frac{TP}{TP+FN}$$

F1_ score: This score will give us the harmonic mean of precision and recall. Mathematically, F1 score is the weighted average of the precision and recall. The best value of F1 would be 1 and worst would be 0.

$(precision * recall)$

$$F1 = 2 * \frac{(precision * recall)}{(precision + recall)}$$

Chapter 2

Literature Survey

2.1 Introduction

The vast and intricate tapestry of life beneath the ocean's surface has captivated humanity for centuries. The ocean covers over 70% of the earth's surface and is known to contain countless species, most of which are yet to be discovered. Therefore, this task is a daunting one that has been lasting for ages. This literature survey aims to delve into the historical and contemporary approaches, methodologies, and technologies employed in the classification and study of underwater species

2.2 Related work

Saleh, et. al [1] survey on various underwater species classification. Deep learning (DL) technology can help aquatic researchers quickly and effectively parse large amounts of video footage, revealing niche details that traditional tracking techniques cannot. From 2003 to 2021 computer visions and DL studies were used to analyse marine environments' fish species. It introduces DL principles, evaluates and integrates these studies, discusses challenges in creating DL for underwater picture handling, and suggests solutions. The paper aims to inform aquatic researchers about the importance of DL principles and the future of DL-based fish categories in their underwater environment.

Ben Tamou A, et. al [2] proposed on various underwater. species classification. They proposed two Deep Convolutional Neural Network (CNN) models for classifying fish species in the unconstrained underwater environment. The initial one is based on the traditional transfer learning framework, combined with a novel technique that is centred around training/validation loss curves to achieve focused data augmentation. On the other hand, hierarchical CNN classification is used by the latter model to classify fish into family levels and species categories. Experiments on two benchmark datasets showed accuracies of 99.86% and 81.53%, respectively, for the proposed approaches.

Mittal, et. al [3] implemented on various underwater species classification. They did a survey on deep learning techniques for underwater image classification, highlighting similarities and differences. It believes underwater image classification is a critical application that tests the success of deep learning techniques. The survey aims to inform
researchers about the latest advancements in deep learning on underwater images and motivate them to advance its frontiers.

Anil, A., et. al [4] proposed on various underwater species classification. They have proposed new deep learning approach for enhancing underwater images. The method involves two stages: segmentation and enhancement. The segmentation stage uses SUIM-Net for semantic segmentation, while the enhancement stage uses a style transfer technique with whitening and colouring transforms. Wavelet pooling preserves structural features while stylizing distorted images. This helps reconstruct signals more accurately. The method's performance is evaluated using metrics like Structural Similarity Index (SSIM), Peak Signal to Noise Ratio (PSNR), Underwater Image Quality Measure (UIQM), and Underwater Colour Image Quality Evaluation (UCIQE). The method achieved better UIQM values compared to other methods.

Ben Tamou, A., et. al [5] highlighted on various underwater species classification. They have proposed a model using a deep Convolutional Neural Network (CNN) and trained this network by using a new strategy based on incremental learning. This strategy involved training the CNN step-by-step, prioritizing recognizing challenging species initially. As training progressed, new species were incrementally added, while ensuring the performance of previously learned species remained high through a knowledge distillation loss. The proposed method achieved an impressive 81.83% accuracy on the Life Clef 2015 Fish benchmark dataset.

Paraschiv, M., et.al [6] suggested on various underwater species classification. They have suggested using small convolutional neural networks for classifying fish images in real-world settings. These networks are efficient and require fewer resources, making them suitable for use in remote fish recognition systems. Even networks with a limited number of parameters (under 12,000) demonstrated good accuracy (around 42% for six fish species) when trained on small and unevenly distributed datasets. The researchers further enhanced accuracy to nearly 49% for six fish species by adding a simple object tracking algorithm to augment the data. These networks have been tested with images obtained from an experimental setup in the Mediterranean Sea.

Domingos, L. C., et.al [7] survey on various underwater species classification. They presents a comprehensive review of deep learning techniques used for automated object categorization in underwater sonar data for monitoring .
classification using passive sonar and object identification using active sonar. They highlighted five key elements in developing these methods: 1. Convolutional layers 2. Biologically inspired filters for feature extraction 3. Data classification using frequency and time-frequency analysis 4. Machine learning methods for feature extraction from raw signals 5. Transfer learning techniques Additionally, they explores data augmentation techniques to overcome the limited availability of labelled sonar datasets collected during actual maritime operations.

Catalan, I. A., et. al [8] survey on various underwater species classification. A new dataset was created to study the impact of labelling small fish on precision, image selection, and the need for random selection. A large dataset was used, containing over 18,400 Mediterranean fish from 20 species, captured in 1,600 underwater images with varying backgrounds. Two detection algorithms, YOLOv5m and Faster RCNN, were evaluated for identifying fish in these images. YOLOv5m outperformed Faster RCNN and was chosen for classifying an increasing number of species using six different combinations of labelled datasets. The study demonstrated that using a smaller set of labelled data is cost-effective, the model's accuracy improves with diverse backgrounds, and avoiding low-quality labels leads to better classification outcomes. These insights can guide

efficient image selection and labelling strategies, enhancing deep learning applications in the study of underwater ecosystems.

Rao, M. C., et.al [9] Proposed on various underwater species classification. Using deep learning, researchers have developed a "machine-assisted" method to speed up analysis of underwater videos from Baited Remote Underwater Video Station (BRUVS) surveys and assist in tracking the health of marine ecosystems. This method automates the process of drawing boxes around fish in the videos, making it faster to label and identify fish species. The system can detect and classify up to 12 different fish species with good accuracy. Using machine learning, researchers developed a tool that can automatically detect and label fish in videos. The tool was accurate, identifying fish with a success rate of 70-89%. It also recognized 12 specific fish species with an F1 score of 0.79. By integrating this tool into video analysis software, fish ecologists could save time and improve their research accuracy.Lopez-Vazquez, V., et. al [10] Analysed on various underwater species classification. A new image enhancement and classification pipeline has been developed for automated processing of images from benthic moving platforms. The pipeline uses a convolutional residual network to generate enhanced images from raw footage. The trained network achieved high underwater imagery assessment metrics, including UIQM and UCIQE. The process also achieved the highest SSIM and PSNR values. The improved images showed promising results, with an accuracy of 66.44% and an AUROC of 82.91%. This strategy is promising for on-board real-time processing of crawler imaging and outperforming previous papers.

Kumar, S., et.al [11] Implemented on various underwater species classification. They introduces an Optimized Deep Neural Network (DNN) framework for developing a Fish Species Classification (FSC) model for underwater image recognition due to noisy captured images. The FSC model uses segmentation and DNN-based identification in complex underwater environments. The model is tested using the Fish4Knowledge (Fish Detection) benchmark dataset, and experimental results show improved performance in terms of Average Precision (AP = 92.26%), Average Recall (AR = 87.45%), Average F-measure

(AF = 89.74%), and Average Accuracy (AA = 93.86%). The composite FSC network model increases the usage of distinctive information and output of distinctive information for the discovered object. The underwater environment is a fascinating and challenging area of research, and researchers are working together to unravel the mysteries of underwater imaging and mapping..

Naseer, A., et. al [12] A detection refinement algorithm based on spatial-temporal analysis has been proposed to improve the performance of generic detectors by suppressing false positives and recovering missed detections in underwater videos. The algorithm uses advanced deep neural networks like Inception, ResNet50, and ResNet101 to automatically classify and detect Norway lobster Nephrops norvegicus burrows from underwater videos. The study used data from the Gulf of Cadiz to evaluate the performance, demonstrating that the proposed framework effectively suppresses false positives and recovers missed detections, resulting in a 10% increase in mean .

Qu, P., et. al [13] highlighted on various underwater species classification. The DAMNet is a deep neural network designed for underwater biological image classification. It uses multi-stage stacking to suppress complex underwater backgrounds, reducing model parameters and improving generalization. The dual attention mechanism module is combined with deep convolution to extract feature information from space and channel aspects, enhancing discrimination and feature extraction. The gravity optimizer updates model weights, and exponential translation improves convergence speed and learning rate. Experiments on seven types of underwater biological images show the DAMNet model has higher learning ability and robustness compared to state-of-the-art methods. It achieves 96.93% classification accuracy in all categories, a 2 percentage point improvement.

Gong, B., et. al [14] analysed on various underwater species classification. Fish-TViT is a novel method for multi-water fish classification based on transfer learning and visual transformers. It uses a label smoothing loss function to address overfitting and overconfidence issues, and Gradient-weighted Category Activation Mapping (Grad- CAM) technology to visualize and understand model features. The method first crops and cleans fish images, then expands the number of training datasets. A pre-trained visual transformer model extracts enhanced features, which are cropped into flat patches. A multi-layer perceptron predicts fish species. Experimental results show Fish- TViT achieves high classification accuracy on low-resolution marine fish data (94.33%) and high-resolution freshwater fish data (98.34%), outperforming traditional convolutional neural networks.

Bilodeau, S. M., et. al [15] developed on various underwater species classification. A new system of low-cost, long-term camera traps (Dispersed Environment Aquatic Cameras) has been developed by them for remote marine environments. These cameras use machine learning to classify the large volume of images collected. A case study was presented to investigate fish movement and feeding behaviour in shallow tropical reef capes. The cameras function continuously underwater at depths up to 40 meters without maintenance .
achieved 92.5% accuracy in sorting images with and without fishes, and diver surveys showed the cameras accurately represented local fish communities. This is the first successful method for broad-scale underwater camera trap deployment.

Deng, Y., et. al [16] Proposed on various underwater species classification. An underwater stereo vision system is used to automatically identify species and measure length using R- To improve the model's robustness, stochastic enhancement is performed on image datasets. An attention module is integrated into the ResNeXt50 network to enhance feature extraction capability. The feature pyramid network (FPN) is replaced by an improved path aggregation network (I-PANet), resulting in a 4.55% and 2.38% increase in object and key point detection tasks. A new algorithm is introduced for matching neural network detection results. The method can quickly and accurately obtain coordinates of head and tail

points in stereo images and fish species. A 3D reconstruction of fish head and tail points is performed using calibration parameters and projection matrix. They proposed an approach has a precision of over 94% for species identification and less than 10% for length measurement. This method can help aquaculture farmers collect real-time information about fish length.

Alaba, S. Y., et. al [17] survey on various underwater species classification. The proposed model uses MobileNetv3-large and VGG16 backbone networks and an SSD detection head to solve the class imbalance problem in a dataset. It uses a class-aware loss function to give more weight to species with fewer instances. This function was applied to any classification or object detection task with imbalanced datasets. Experimental results show that the class-aware loss improves the model by up to 79.7% on the large-scale reef fish dataset, SEAMAPD21, and outperforms the original SSD object detection model.

Mol, J. J., et. al [18] survey on various underwater species classification. They proposed an enhanced grey wolf optimization technique (EGWO) for fish classification using a deep learning-based auto encoder decoder method. The EGWO efficiently selects the most optimal texture features from the auto encoder extracted features. The accuracy of the classification system depends on essential texture features, and the proposed method is compared to existing deep learning models like Alex Net, Res Net, VGG Net, and CNN. The performance criteria such as accuracy, sensitivity, specificity, precision, and F1 score show that AED-EGWO provides superior performance.

Zhai, J., et. al [19] Suggested on various underwater species classification. They proposed the Sandwich Attention Covariance Metric Network (SACovaMNet), a novel attention network that enhances the CNN's ability to extract features on fine-grained fish images. The new model captures global classification objects and extracts local subtle differences, enabling accurate classification of marine fish images. Experiments show that this method outperforms state-of-the-art solutions

in few-shot fine-grained fish species classification, demonstrating its potential for feature enhancement.

Desai, N. P., et .al [20] Suggested on various underwater species classification. They utilized advanced computer vision and data mining techniques to find the best model, despite limited research scope and difficulties. They compared it to state-of-the-art models like CNN and Efficient Net, and found the DNN model to be the leading model with 100% accuracy.

Li, Y., et. al [21] Analysed on various underwater species classification. They used YOLOV5 underwater object detection system uses an attention mechanism and multiple-scale detection strategies to identify four common marine organisms in natural scenes. An image enhancement module enhances image quality and observation range, while a triplet attention mechanism enhances feature extraction. The prediction head structure is optimized for small-sized objects. Ablation studies validate the performance of each module. The model outperforms state-of-the-art models in accuracy and speed. It was deployed on an embedded device with a processing time of less than 1 second, making it suitable for real-time observation by mobile platforms or underwater equipment.

Wulandari, N., et. al [22] trained five models on the RUIE dataset. The study compared the average detection time and accuracy of five underwater object detection models.

SSD was fast but had poor performance, Retina Net had consistent performance but slow detection speed, and YOLOv3 was the fastest and most acceptable. YOLOv4 was good initially but dropped performance as threshold enlargement increased. Despite no fully suited models for underwater object detection, YOLOv3 was found to be the best acceptable model among the evaluated models, with YOLOv3 being the most accurate.

Ahmad, U., et. al [23] They proposed a transfer learning-based technique using Efficient-Net, pre-trained on the ImageNet dataset and fine-tuned on the QuT Fish Database. The model uses Global Average Pooling (GAP) instead of a dense layer to average predictions and provide more information. The model was validated on a validation set, achieving satisfactory results. The localization task is addressed using a localization aware block and residual connections to handle over-fitting and class imbalance problems. Class weights and Focal Loss (FL) are used to reduce false predictions. The qualitative assessment shows a 57% Mean Intersection Over Union (IoU) on testing data, with classification results showing 75% precision, 70% recall, 78% accuracy, and 74% F1-Score for 468 fish species.

Volkan, K. A. Y. A., et.al [24] proposed model The IsVoNet8 model, when compared with ResNet50, ResNet101, and VGG16 models, achieved success accuracies of 98.62% and 91.37% respectively. However, the loss rates of ResNet50, ResNet101, and VGG16 were significantly lower than IsVoNet, indicating that the IsVoNet effectively classifies marine fish, a popular food source in Türkiye.

Gupta, A., et. al [25] They introduced the fish detector YOLO Fish, which uses hierarchical techniques to detect underwater fish species. This method addresses low visibility and makes it easier to distinguish features. With a 91.8% map accuracy, YOLO Fish is a state-of-the-art object detector for Nordic fish species. The algorithm has an inference time of 26.4 Ms, making it fast enough for real-time video on a high- end GPU Tesla V100.

Muri, K. K., et. al [26] A two-step deep learning approach has been proposed for detecting and classifying temperate fishes without pre-filtering. The first step involves detecting each fish in an image using the You Only Look Once (YOLO) object detection technique. The second step uses a Convolutional Neural Network (CNN) with the Squeeze-and-Excitation (SE) architecture for classifying each fish without pre- filtering. Transfer learning is applied to improve the accuracy of classification. The model is trained using ImageNet and the fish classifier via a public dataset (Fish4Knowledge), with weights applied to post-training. The solution achieves a state- of-the-art accuracy of 99.27% using the pre-training model and 83.68% and 87.74% with and without image augmentation, respectively.

Hassoon, I. M. et. al [27]). Suggested on various underwater species classification. They compares fish identification techniques based on pre-processing, significant characteristics, and correctness. Fish are crucial for human food and economic growth, and understanding their benefits and diseases is essential for building a knowledge base. The paper reviews various fish species identification techniques, including pre- processing operations, features extraction methods, recognition/classification techniques, and databases. It is beneficial for beginner researchers and serves as a starting point for new fish species identification approaches. The review aims to provide a comprehensive understanding of fish species.

Kalaiarasi, G., et .al [28] Analysed on various underwater species classification. The purpose of this paper is to present a systematic method for analysing recent underwater pipeline imagery using deep learning. There is a logical organization of the analytical methods based on the recognized items, as well as an outline of the deep learning architectures employed. Deep neural network analysis of digital photographs of the seafloor has a lot of potential for automation, particularly in the discovery and monitoring of underwater pipeline images.

Iakushkin, O. O., et. al [29] survey on various underwater species classification. They proposed a new system to analyse underwater video data, reducing time spent on analysis. The system divides video into frames, assesses noise and blurriness, corrects

for resolution increase, analyses animal count, and reports on the content. The data is displayed on a map, allowing for precise area mapping and surveillance. The system also calculates biotope mass by correlated Few-shot learning segmentation model results with point cloud data, providing a biotope surface coverage area. This system uses automatic processing of a single camera underwater video data and combines deep neural networks into a single pipeline.

Hong Khai, T., et.al [30] Proposed on various underwater species classification. They trained a model using a robotic eye camera to capture shrimp photos from a shrimp farm. The data was classified into low, medium, and high- density categories. The model was improved using parameter calibration, achieving an accuracy rate of up to 97.48%. The model's effectiveness was further enhanced.

Denny, C. M, et. al [31] Developed on underwater species classification. The Marine Park in the region has been found to have the lowest mean numbers and sizes of snapper compared to other areas, including the fully no-take Poor Knights Island Marine Reserve and two other reference areas open to all kinds of fishing. The Marine Park also had the lowest mean numbers and sizes of pigfish, leatherjackets, and trevally, but none of these species are heavily targeted by fishers. The underwater visual census showed five species significantly more abundant in the Marine Park and five species more abundant outside the park. The lack of recovery by snapper within the Marine Park, despite restrictions on recreational fishing, suggests that partial closures are ineffective as conservation tools. The data suggests that fishing pressure within the Marine Park is at least as high as at other "fished" sites

Jiang, X.,, et. al [32] developed on underwater species classification. They presents a marine organism detection framework using a dataset augmentation strategy and a Convolutional Neural Networks (CNN)-Vision Transformer (ViT) fusion model. The framework uses random expansion of small objects and non-overlapping filling of scarce samples to improve dataset quality. It uses YOLOv5 as the baseline model and introduces ViT, deformable convolution, and trident block in the feature extraction network. The experimental results show a 27% improvement in mean average precision
compared to various one-stage detection models. The framework also considers performance and real-time for high-precision real-time detection of marine organisms on underwater mobile platforms.

Heidari, et.al [33] developed on underwater species classification. A field experiment at the University of Zabol in Iran revealed that water stress significantly altered antioxidant activity in basil plants. The highest concentrations of CAT and GPX activity were found in W3 treatments. Chlorophyll content in leaves increased with increasing water stress, while Fv/Fm and APX activity decreased. Rhizobacteria application improved antioxidant and photosynthetic pigments in basil plants. Pseudomonades sp. increased CAT enzyme activity under water stress, while S4 (combination of three bacterial species) had the highest GPX and APX activity and chlorophyll content in leaves.

Almubarak, et. al [34] suggested on underwater species classification. They have proposed an underwater robot called Kraken, which mimics the arm movement of an octopus using a hybrid actuation technology of stepper motors and twisted and coiled polymer fishing line muscle (TCPFL). The tentacles are designed to operate in various modalities, such as curling, twisting, and bending, to control their shape and grasp irregular objects. Kraken also features an onboard battery, wireless programmable joystick, and buoyancy system for depth control. The robot is fully

functional underwater in an Olympic-sized swimming pool, and the researchers have proposed an embedded TCPFL actuator within elastomer for the tentacles, enhancing the robot's performance.

Huy, et. al [35] Analysed on underwater species classification. Underwater perception faces challenges like rapid light attenuation, light refraction, and back-scattering, reducing the signal-to-noise ratio (SNR) of sensing devices. This paper discusses state- of-the-art sensing technologies and object perception techniques for underwater robots in different environmental conditions. It divides perception ranges into close-range, medium-range, and long-range, describing recent advances and suggesting potential future research directions in this field.

Toyouchi, et. al [36] Proposed on underwater species classification. A laser on graphene under water can introduce a sp3-type chemical defect through photo-oxidation, which was reverted to sp2 carbon centres using the same laser with higher intensity. This photoreduction occurs due to laser-induced local heating on the graphene. Combining optical methods with a laser direct writing technique allows photo writing and erasing of a well-defined chemical pattern on a graphene canvas with a spatial resolution of about 300 nm. Raman mapping visualizes the pattern, enabling an optical read-out of chemical information. This all-optical method allows flexible and efficient tailoring of nanoscale physicochemical properties for future applications.

Jung, et. al [37] Implemented on underwater species classification. Thermal gradient energy-generation technologies for powering unmanned underwater vehicles (UUVs) or autonomous sensing systems in the ocean are mainly in the research development phase or commercially available at a limited scale. Salinity-gradient energy-generation technologies have not been adequately researched yet. The demand for self-powered UUVs suitable for long-term deployments is growing, and further research is needed on small-scale ocean gradient energy

systems. This study reviews harvesting energy from ocean thermal or salinity gradients for UUVs, focusing on gliders and profiling floats. Thermal gradient energy systems based on phase change materials (PCM) cannot provide the energy required for autonomous sensing systems due to low energy conversion efficiency. Developing more efficient electrical-mechanical systems and enhancing thermal conductivity of PCMs may help address this challenge. Emerging technologies like thermoelectric generators, shape memory alloys, and small-scale thermodynamic cycle systems have potential for UUV power, but are still in the laboratory testing or conceptual design phase.

Srivastava, et. al [38] Suggested on underwater species classification. They investigates the impact of water shortage on Brassica juncea plants, also known as Indian mustard. They tested eight different treatments under different conditions, including control, stress, and stress conditions. They evaluated 16 traits, including plant response to stress, peroxidase (POD), superoxide dismutase (SOD), and catalase (CAT) activity. The

results showed that fresh weight increased only 7.47 percent with mycorrhiza and 22.39 percent with silicon, but 291.08 percent with both mycorrhiza and silicon. Using mycorrhiza or silicon alone resulted in a significant increase of 53.16 percent and 55.84 percent in fresh weight, respectively. However, using both mycorrhiza and silicon together resulted in a dramatic increase of 380.71 percent under stress conditions. Superoxidase dismutase concentration increased by 29.48 percent, 6.71 percent, and
22.63 percent after applying C+My, C+Si, and C+Si+My. Carotenoid content increased by 58.06 percent, 54.83 percent, 183.87 percent with C+My, and 23.81 percent with S+My and S+Si+My, but decreased by 22.22 percent with S+Si. Silicon application proved more effective than AMF treatment with Rhizophagus irregulars, and the best results were obtained with the combination of Si and AMF.

Dukamo, et. al [39] Analysed on underwater species classification. The study aimed to analyse the Geno-typic diversity of Ethiopian durum wheat genetic resources under two contrasting environments: drought-stressed and non-stressed. It involved 100 landraces and 4 local checks, collecting data on 13 traits including yield, yield components, phenology, and canopy condition. The analysis revealed significant differences between landraces for different traits with different sources of variation. Several landraces outperformed the checks at both environmental conditions. The study found intermediate to high estimates of phenotypic coefficient of variation (PCV), genotypic coefficient of variation (GCV), heritability in a broad sense (h2b), and genetic advance in percent of the mean (GAPM) for all studied traits except days to flowering, thousands seed weight at stress, days to maturity, leaf chlorophyll concentration measurement, and canopy temperature measurement. Genotypic variation was found to be higher than environmental variation for most traits. The number of tillers, spike length, kernel per spike, and grain yield showed higher values for h2b and GAPM at normal and stressed conditions, respectively. Spikelet's per spike, kernel per spike, and thousands seed weight were positively correlated with grain yield, while grain yield exhibited a weak negative correlation with days to heading and days to maturity. Cluster analysis grouped the landraces into six clusters, with each showing variation in performance for different traits under normal and stressed conditions. The presence of significant genetic variability among the evaluated durum wheat landraces suggests an opportunity for improving grain yield through hybridization of genotypes from different clusters.

Alsmadi, et. al [40] Proposed on underwater species classification. Fish classification (FC) techniques are widely used in various fields, including target marketing, to maintain fish supply and balance between ecosystems, commercial, agriculture, marine scientists, and industrial sectors. This survey compares the performance of various FC techniques based on preprocessing and feature extraction methods, classification accuracy, and recognized fish families/species. Databases like Fish4-Knowledge, knowledge database, and GIS on fishes are reviewed. The study aims to enhance understanding of preprocessing methods,

features extraction techniques, and classifiers, guiding future research and filling current research gaps.

Fernandez Garcia, et. al [41] Suggested on underwater species classification. They have developed a novel approach using convolutional neural network (CNN) and classical computer vision (CV) techniques to detect fish passages in acoustic video streams. The pipeline pre-treats the images to localize the signals of interest and improve detection performances. The YOLOv3-based model was trained with fish data from multiple species recorded by the DIDSON and ARIS cameras, including species of high ecological interest like Atlantic salmon and European eels. The pre-treatment of images significantly improves the model's performance, increasing its F1-score from 0.52 to
0.69. The model detects almost 80% of fish passages and minimizes false-positive rates. Efficiency increases with fish sizes, reaching a recall higher than 95% for Atlantic salmon. However, the model appears less efficient for eel detections on ARIS videos than on DIDSON data.

Risholm, et. al [42] Developed on underwater species classification. They proposed an underwater 3D range-gated camera for precise fish-length estimation of free-swimming fish. The system, which doesn't require manual work or contact with the fish, uses a robust pipeline for detection, tracking, and estimation stages. The system achieves a 1% accuracy rate in length estimation, demonstrating its effectiveness in both individual and population distributions of free-swimming fish.

Duan, et. al [43] Suggested on underwater species classification. They investigates the capacity to restore hydraulic function post-drought in six tree species, including evergreen and deciduous, and its implications for future forest composition. They found differences in drought responses and recovery between species, but not as a function of evergreen or deciduous groups. Sapindus mukorossi exhibited the most rapid drought response, associated with physiological and morphological traits. Upon re-watering, xylem water potential

exhibited fast recovery in 1-3 days among species, while photosynthesis at saturating light (Asat) and stomatal conductance (gs) recovery lagged behind water potential recovery depending on species. None of the six species exhibited significant hydraulic recovery during the 7-day re-watering period, indicating that xylem refilling was limited. Additionally, NSC availability had a minimal role in facilitating hydraulic recovery during this short-term period. They suggests that if water supply is limited by insignificant hydraulic recovery post-drought, they observed carbon assimilation recovery of seedlings may not be sustained over the longer term, potentially altering seedling regeneration and shifting forest species composition in subtropical China under climate change.

Ricca, et. al [44] Implemented on underwater species classification. They evaluated the mechanical and antifouling properties of two underwater mortars containing Mg(OH)2 as an antifouling agent. The mortars' mechanical properties were assessed through uniaxial compressive strength and flexural strength measurements. The materials' composition was explored using DTA-TG, XRPD, SEM, and EDS microanalysis. They found that both mortars had good mechanical resistance when set underwater, and the addition of Mg(OH)2 improved resistance against biofouling, observed in both laboratory and sea-exposed specimens.

Liconti, et. al [45] survey on underwater species classification. Gorgonian forests, a complex subtidal habitat in the Mediterranean Sea, are crucial for biodiversity and ecosystem services. However, their geographical distribution and condition are poorly understood. A study aimed to assess the biological complexity of gorgonian forests in Italian coastal waters, evaluate impacts and vulnerable species, and identify areas of special interest within and outside marine protected areas to prioritize conservation strategies.

Amanda Lucas Pereira, et. al [46] Analysed on various underwater species classification. They proposed using image classification models to assist

underwater inspection specialists in identifying relevant events. However, challenges like high costs, lack of balanced data, label noise, and underwater image characteristics like turbidity and uneven lightning conditions make traditional supervised models insufficient. They propose DINO and mPAWS, a multi-label version of PAWS, to address these issues. The results show a $2.7% improvement in performance compared to current supervised models, and real-time inference can accelerate underwater vehicle tasks, making specialists' work more efficient.

Mengwei Li, et. al [47] Developed on various underwater species. They proposed a fish recognition method based on the improved YOLOv5 model to improve the speed and accuracy of detecting marine fish species in natural environments. The method incorporates an efficient channel attention mechanism module to enhance local and global feature learning and modifies the GIoU loss function to reduce the impact of low-quality samples on detection. Under the same training conditions, the improved YOLOv5 model achieves 94.6% detection precision, 93.8% recall rate, and 92.4% mean average precision, making it suitable for detecting marine fish species in natural environments.

Wanxin Liang, et. al [48] Suggested on various underwater species. A super-resolution underwater image segmentation network was proposed using multi-scale fusion and Resnet50 as the backbone feature extraction network. This enhanced feature capture capability and refined output was achieved through up sampling output and the super- resolution module. The model demonstrated significant segmentation capability on the SUIM underwater image segmentation dataset, with better shallow feature restoration and preservation, higher accuracy of edge feature information, and higher cross-merge ratio and accuracy. The network outperformed other networks on the SUIM dataset,
with improved average cross-merge ratio and pixel accuracy. The network's performance was compared with other networks on the same dataset.

Braga, L. T. M. D., et. al [49] Proposed on various underwater species. They used a Dual frequency Identification SONar (DIDSON) to record and count fish in a dam using an innovative approach. They compared three methods of counting fish manually in DIDSON software frames from two Brazilian dams: numerical counting, qualitative counting, and intersection counting. They found that numerical counting provided the best estimates of true value but could not be used in 32.2% of frames due to uncountable aggregation of fish, bubble curtain, or poor contrast between fish and the dam floor. Qualitative counting was the most congruent, while intersection counting was the least.

Buenau, et. al [50] Implemented on various underwater species. They examines the impact of tidal, wave, and ocean current energy converters on marine ecosystems. It highlights the availability and maturity of models for stressor-receptor interactions and provides examples involving ME devices. Models for oceanographic systems and underwater noise are widely available but require validation in real-world settings. Few examples of habitat change and displacement of marine animals related to ME exist. Models for collision risk and species response to EMFs are still in development and require more observational data. They concludes by synthesizing model status, commonalities, and overlapping monitoring needs for efficient environmental impact prediction and monitoring.

Y. Luo, X. Xiao, et. al [51] Analysed on various underwater species. A bioinspired soft sensor array (BOSSA) has been proposed to integrate pressure and material sensing capabilities based on the triboelectric effect. The array uses cascaded row + column electrodes embedded in low-modulus porous silicone rubber to capture rich information from the environment. Data-driven algorithms extract higher-level features. BOSSA can identify 10 users (98.9%) and 10 objects (98.6%). Its scalable fabrication allows for large-area sensor arrays with high spatiotemporal resolution and multimodal sensing abilities, yet with a less complex system architecture. These features was beneficial for intelligent monitoring and stimuli response in smart home and industry applications.

E. Marotte, et. al [52] survey on various underwater species. The North Atlantic right whale (NARW) is under threat from anthropogenic underwater noise, which is causing significant stress. The Canadian government is working to understand the sources of noise and their impacts, but there is a lack of knowledge about which noise sources are most affecting NARW in their habitat. They presents the results of a 2020 workshop aimed at developing specific metrics to better characterize noise sources and their impacts on NARW. The recommendations aim to provide a basis for more targeted research and effective management and protection of NARW, potentially extending to similar species.

S.W.K. van den Burg, et. al [53] Developed on various underwater species. They examines the practices in monitoring and evaluating maritime spatial plans, based on literature, interviews, and expert workshops. It highlights the need for more than just identifying best methods; it requires sufficient skills and resources for evaluators and stakeholders. Equity in marine area allocation is a growing issue, and linear approaches to monitoring and evaluation fail to capture this complexity. The value of M&E lies in promoting a participative process with cross-sectoral and cross-border learning, which can improve the quality of maritime spatial plans.

Cheng, M. L. H., et. al [54] Proposed on various underwater species. An acoustic imaging microtome system (AIMS) has been proposed to map the internal structure of fish. The system consists of two pairs of high-frequency planar arrays, providing a series of acoustic images along the fish's length. The system measures acoustic attenuation along the direct ray paths of each transmit-receive channel pair, resulting in a 2-D image of the fish's interior morphology with 4- to 5-mm resolution. The images provide information on the fish's cheekbone, backbone, rib bone, skull, fins, stomach, and swim bladder, which is crucial for accurately modelling fish acoustic target strength, especially for non-swim bladder-bearing species.

T. Kim, et. al [55] survey on various underwater species. They used a mixed ion-electron conductor to generate a potential spike signal, creating artificial receptors with different ion relaxation times. This creates a spike spectrum, providing spatial and

temporal information. The device was incorporated up to 132/square centimetres using only two global signal addressing lines. The device's structural simplicity allows for scalable fabrication with dense receptor integration. The computational decoding of the position-encoded spike spectrum enables real-time recognition of complex dynamic motions, enabling real-time dynamic robotic manipulation.

Y. Cheng, et. al [56] survey on various underwater species classification. Aquatic skin is an underwater sensing strategy that incorporates contact pressure, tactile mapping, depth, temperature, and salinity in a flexible architecture. This innovative approach simplifies sensor design, material preparation, device fabrication, and signal processing. The inverse inotropic sensing mechanism uses elastomeric-electrode and environmental-electrode interfaces for contact pressure detection, achieving exceptional resolution and hydraulic balance. A hydrophobic ionic gel with adjustable surface morphologies serves as the functional sensing layer, ensuring long-term stability. The aquatic skin can achieve sub-Pascal resolution for contact pressure detection and sub-millimetre spatial resolution for tactile mapping over a wide depth range. It is used for real-time monitoring of vital/environmental signals, tactile recognition of creatures, and biomechanical analysis of fish swimming.

Ji, W., et. al [57] Analysed on various underwater species classification. They proposes a multi-scale pyramid fusion method to improve underwater crab images, enhancing contrast and removing fog. It also presents a crab target detection method using the Mobile Centre Net model, which extracts crab features using an improved MobileNetv2 backbone network with coordinate attention module. The Feature Fusion Module (FFM) extracts multi-scale feature map

information, and Atreus Spatial Pyramid Pooling (ASPP) fuses context information from different receptive fields. The experimental results show that the Mobile Centre Net model achieves 97.86% and 97.94% average precision and F1 values, with a model size of 24.46 M and a detection speed of 48.18 frames/s. The model also reduces storage memory required for model training by 81% and increases AP on the crab dataset by 3.2%. The method is effective for real-time monitoring and scientific feeding in crab breeding

Wu, G. K., et. al [58] Proposed on various underwater species classification. They proposes an enhancement computing method for underwater microscopic images of High-Altitude Biological Species (HABs). The method uses cyclic colour compensation to correct the colour of the images, and then redefines the correlation weight maps of the three inputs to transfer texture features and colour contrast to the output images. A novel image multi-scale fusion strategy is used to avoid image artifacts in low-frequency components. The method improves the accuracy of transmission map estimation, edge detection, and key point matching in processing algal images, resulting in better texture clarity and global contrast.

Böer, G., et. al [59] Implemented on various underwater species classification. They presents an end-to-end processing pipeline that measures the number and type of biological species and estimates their size directly from stereo video capture by the stereo camera on Underwater Fish Observatory (UFO) stations. The system was calibrated in situ and validated by simultaneous recording with sonar. The raw data were acquired continuously in Kiel Fjord, Germany, for almost a year. The recorded raw data are pre-filtered by an adaptive background estimation to extract sequences with activity, which are processed by a deep detection network, Yolov5. This provides the location and type of organisms detected in each video frame, which are used to calculate stereo correspondences. The size and distance of depicted organisms are approximated using the corner coordinates of matched bounding boxes. The Yolov5 model was trained on a novel dataset of 73,144 images and 92,899 bounding box annotations for 10 categories of marine animals.

Yang, J., et. al [60] survey on various underwater species classification. DARE is a diver action recognition encoder that can identify multiple diver actions underwater, distinguishing sixteen gestures and three body positions. It uses two bi-channel conjugate convolutional layers for feature extraction on underwater camera images and is trained on the Cognitive Autonomous Diving Buddy (CADDY) dataset. DARE takes only a few milliseconds to classify each stereo-pair, making it suitable for real-time applications. It achieves a 95.87% overall accuracy and 92% minimum class accuracy, confirming its robustness and reliability. DARE also improves the performance of class
baseline classifiers by 3.44% in accuracy and 30% in minimum class accuracy, proving its robustness and reliability.

2.3 Research Gap

The biodiversity of deep-sea environments is poorly understood, with a significant gap in knowledge regarding species, ecological roles, and adaptations. While molecular techniques like DNA barcoding and eDNA analysis have revolutionized species identification, there is a need for greater standardization and validation. Combining traditional morphological data with molecular, ecological, and environmental data can provide a more comprehensive understanding of species, but effective methods for integrating and analysing these diverse data types are still underdeveloped.

Climate change is affecting marine species, particularly in terms of species distribution, migration patterns, and adaptation. Research in this area is crucial for effective conservation and management strategies. Advances in AI and machine learning for species identification and monitoring are still lacking, and there is a lack of long-term, continuous ecological studies.

More research is needed on the impact of invasive species on native marine ecosystems, their long-term ecological consequences, and effective management strategies. The diversity and role of marine microbes are not as well understood as larger marine organisms, and more effective methods for prioritizing conservation efforts are needed.

Understanding the socio-economic impacts of changes in marine biodiversity is also needed. Addressing these research gaps requires interdisciplinary approaches incorporating marine biology, ecology, genetics, oceanography, computer science, and social sciences.

2.4 Comparative analysis

REF	Work Done/ Methodology	Observations/ Drawback's	outcome
[1]	Visual AI YOLOv5 algorithm for object detection. AI learning classification model (YOLO-Classification) for species identification.	Small number of target species for classification accuracy. Need for high-quality image database for accurate species classification.	
[2]	Transfer learning with targeted data augmentation. Hierarchical CNN classification for fish species identification.		
[3]	Survey of deep learning techniques for underwater image classification. Comparison of similarities and differences of several methods.	The paper provides a survey of deep learning techniques for underwater image classification, highlighting similarities and differences among	

		various methods.	
[4]	Style transfer technique with encoder-decoder circuit. Semantic segmentation using SUIM-Net	They proposes a novel style transfer based deep learning approach for enhancing underwater images using a two-stage process: segmentation and enhancement. Various metrics are used to evaluate the performance of the method.	
[5]	Deep Convolutional Neural Network (CNN). Incremental learning with knowledge distillation loss.	The paper proposes a method for live fish species classification in underwater images using a deep Convolutional Neural Network (CNN) trained with incremental learning and knowledge distillation loss	Accuracy=81.8%
[6]	Small convolutional neural networks for fish image classification. Object tracking algorithm for data augmentation.	Limited number of underwater fish images and videos available. Presence of similar fish individuals and limited variety in the dataset.	

[7]	Deep-learning methods using convolutional layers alone. Deep-learning methods that apply biologically inspired feature-extraction filters as a preprocessing step.	Scarcity of annotated sonar datasets from real maritime missions. Lack of dedicated acoustic feature-extraction strategies for audio signals.	
[8]	Two state-of-the-art object detectors/classifiers, YOLOv5m and Faster RCNN, were compared. Different datasets were created using underwater images from various sources.	The paper discusses the automatic detection and classification of coastal Mediterranean fish from underwater images using deep learning techniques.	
[10]	Contrast Limited Adaptive Histogram Equalization (CLAHE). Gamma Correction. Colour Balance. Convolutional Neural Networks (CNNs). Autoencoders.	Strong environmental variability in marine environments requires image pre-processing. Manual processing of large datasets is infeasible, necessitating automated image enhancement.	
[11]	Optimized DNN framework called Fish Species Classification	Existing models faced issues in recognizing fish species from	Precision=92.26% Recall = 87.45%

	(FSC) model. Fish detection based on segmentation approach and DNN-based identification.	underwater images. Developing a high-performance fish recognition model is challenging due to the chaotic nature of underwater imagery.	
[12]	Object detection using state-of-the-art generic detectors. Detection refinement using spatial-temporal filtering (STF) to recover missed detections and suppress false positive.	Challenges in underwater object detection: image blurring, texture distortion, colour shift, scale variation. Low precision and recall rates of generic CNN-based object detectors.	
[13]	Multi-stage stacking to suppress complex underwater background. Dual attention mechanism module combined with improved reverse residual bottleneck.	The DAMNet model uses a dual attention mechanism and stacking to suppress complex underwater backgrounds and improve feature extraction, achieving higher accuracy in underwater biological image classification.	
[14]	You Only Look Once (YOLO) architecture-based method. Deep neural fish	Underwater classification challenges: background noises, image distortion,	Accuracy= 99.89%

	classification system.	occlusion. Existing algorithms focus on classifying fishes outside of water.	
[15]	Low-cost, long-term camera traps (Dispersed Environment Aquatic Cameras). Machine learning classification using ResNet-50-based deep learning model.	Limitations of the aquatic environment for deploying camera traps in marine systems. Biofouling impairing image quality after 1-2 months of camera operation.	Accuracy= 92.5%
[16]	R-CNN approach for species identification and length measurement. Integration of attention module and improved path aggregation network (I-PANet) for feature extraction.	The paper presents an automatic recognition method for fish species and length using an underwater stereo vision system. It does not specifically mention the word "fishes" in the context of the query.	Accuracy = 94%
[17]	The proposed model consists of MobileNetv3-large and VGG16 backbone networks and an SSD detection head. A class-aware loss function is proposed to handle the imbalanced	Information loss affects small objects due to pooling operation in deep networks. Missed detections in some cases, especially using the MobileNetv3 backbone.	Loss =79.7%.

	dataset.		
[19]	Sandwich Attention Covariance Metric Network (SACovaMNet). CovaMNet based on metric learning.	The paper proposes a method called SACovaMNet for few-shot fine-grained fish species classification, addressing the challenges of limited data and classification difficulties in the marine environment.	
[21]	Global attention mechanism (GAM) added to the backbone network. Fusion block based on DAMO-YOLO introduced for the neck.	The proposed model achieved an mAP@0.5 score of 80.2% on the URPC2019 dataset. The model's mAP@0.5:0.95 reached 51.0% on the MS COCO dataset.	
[23]	Transfer learning with Efficient-Net and QuT Fish Database. Localization aware block with residual connections	Lack of data imbalance problems and large number of classes. Over-fitting problem in the localization task.	
[25]	Hierarchical structure-based YOLO Fish algorithm. Use of hierarchical classification and other techniques.	Underwater conditions make it difficult to discriminate fish species.	Precision = 91.8%

[27]	Pre-processing operations, features extraction methods, recognition/classification techniques, and databases are discussed. The paper provides a review of various fish species identification techniques.	Fish species identification process is problematical and time-consuming. Automated fish identification is important due to these limitations.	
[28]	Combination of max-RGB and shades of grey methods. RCNN (Region-based Convolutional Neural Network) technique.	Short detection ranges and poor visibility in underwater locations. Poorly illuminated underwater images.	
[29]	The paper presents a tool that uses AI to analyse underwater video data. The tool uses a pipeline of deep neural networks for video analysis.	The paper presents a tool that reduces the time spent on video analysis. The paper presents a system for detailed underwater biotope mapping.	
[30]	The paper uses Mask R-CNN and an improved Mask R-CNN model. The paper also uses a parameter calibration strategy.	The improved Mask R-CNN model achieved an accuracy rate of up to 97.48%. The precision and recall of the improved model showed	Acc = 97.48%

		significant improvement compared to the existing Mask R-CNN model	
[33]	Field experiment conducted at the University of Zabol in Iran Split plot design with three levels of water stress and four levels of bacterial strain.	Water stress and bacterial strain affected proline and soluble carbohydrate accumulations. Chlorophyll content increased in all bacterial strain treatments.	
[36]	All-optical method for spatially resolved and reversible chemical modification of graphene. Laser direct writing technique for photowriting and erasing chemical patterns on graphene.	A laser can introduce and reverse chemical modifications on graphene. The method allows for photo writing and erasing of chemical patterns on graphene.	

CHAPTER 3

EXISTING METHODOLOGY

3.1 Support Vector Machine

Support Vector Machine (SVM) is a powerful machine learning algorithm used for linear or nonlinear classification, regression, or even outlier detection responsibilities. SVMs was used for quite a few obligations, which include text category, picture classification, junk mail detection, handwriting identity, gene expression evaluation, face detection, and anomaly detection. SVMs are adaptable and efficient in a number of programs because they could manage excessive-dimensional information and nonlinear relationships.

SVM chooses the intense factors/vectors that assist in growing the hyperplane. These intense instances are known as support vectors, and hence algorithm is termed as Support Vector Machine. Consider the underneath diagram wherein there are two distinct categories which are categorized the usage of a choice boundary or hyperplane:

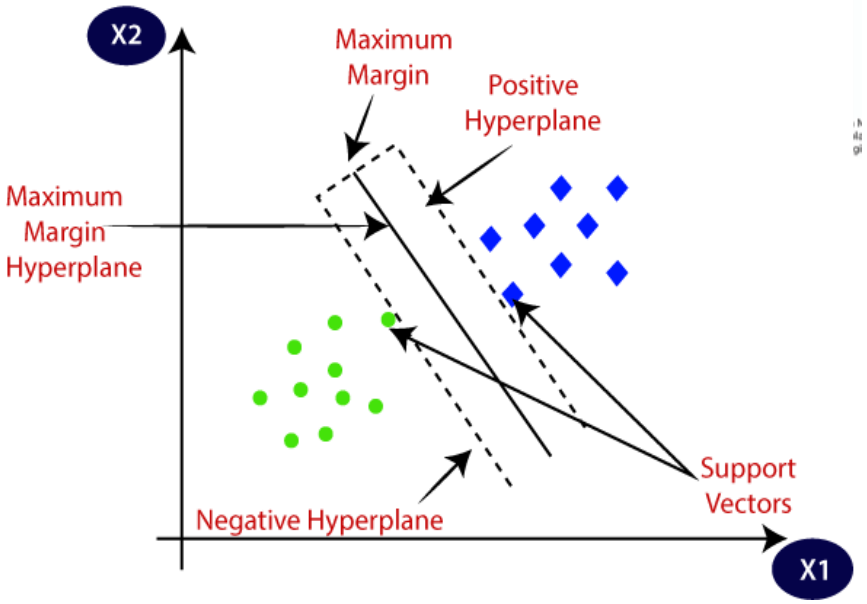


Fig 3.1 Support Vector Machine

The main idea behind SVM is to find the optimal hyperplane that best separates the data points into different classes. The optimal hyperplane is the one that maximizes the margin between the nearest points of the classes, which are known as support vectors.

This concept is primarily applied in a binary classification setting, but SVM was extended to multiclass classification through strategies like one-vs-one and one-vs-all.

The goal of the SVM algorithm is to create the satisfactory line or selection boundary that could segregate n-dimensional space into instructions in order that we can without problems put the new records factor in the precise class in the future. This fine decision boundary is referred to as a hyperplane.

SVM Terminology :

- **Support Vectors** - The data points that lie closest to the decision boundary (hyperplane) and influence its position and orientation.
- **Margins** - The distance between the hyperplane and the support vectors. The goal of SVM is to maximize the margins.
- **Hyperplane -** The decision boundary that separates the classes. Maximum margin hyperplanes are optimal separators.
- **Kernel -** A function that transforms the input data into a different feature space to make it more separable. Common kernels are linear, polynomial, radial basis function (RBF), etc.

Types of Support Vector Machines

- **Linear SVM:** Linear SVMs use a linear decision boundary to separate the data points of different classes. When the data was precisely linearly separated, linear SVMs are very suitable. This means that a single straight line (in 2D) or a hyperplane (in higher dimensions) can entirely divide the data points into their respective classes. A hyperplane that maximizes the margin between the classes is the decision boundary.
- **Non-Linear SVM:** Non-Linear SVM was used to classify data when it cannot be separated into two classes by a straight line (in the case of 2D). By using kernel functions, nonlinear SVMs can handle nonlinearly separable data. The original input data is transformed by these kernel functions into a higher- dimensional feature space, where the data points was linearly separated. A linear SVM is used to locate a nonlinear decision boundary in this modified space.

3.2 Existing Drawbacks

- Prone to overfitting
- Only handles binary classification
- Difficulty handling unknown data
- Fixed structure
- No probability estimates

CHAPTER 4

PROPOSED METHODOLOGY

4.1 Overview

Convolutional Neural Networks (CNNs) have been widely employed in the classification of underwater species due to their ability to effectively learn spatial hierarchies of features from images. CNNs in the classification of underwater species has numerous practical applications across various domains like aquaculture and fishers' management, underwater robotics and autonomous systems, environmental monitoring and pollution control, scientific research and education, recreational diving.

In the realm of underwater species classification, a comprehensive workflow encompasses data collection, annotation, preprocessing, model architecture selection, training, evaluation, and deployment. Data collection involves gathering images using various tools like underwater cameras, ROVs, or AUVs across diverse underwater environments. These images are annotated with species labels, either manually by experts or through automated tools. Preprocessing steps ensure data quality and uniformity, including resizing, normalization, augmentation, and background removal. Model architecture selection involves choosing from established CNN architectures like VGG or Inception, or custom designs tailored to the task. Training the model involves feeding labelled datasets through the network, optimizing parameters using algorithms like SGD or Adam. Transfer learning from pre-trained models was employed to mitigate data scarcity. Evaluation assesses model performance using metrics like accuracy and precision, with testing on unseen data. Finally, deployment involves packaging the model for real-world use, ensuring compatibility and reliability in operational settings, and maintaining the model's effectiveness over time through monitoring and updates. This comprehensive overview highlights the iterative nature of developing and deploying effective underwater species classification systems, ultimately contributing to scientific research and conservation efforts in underwater ecosystems.

4.2 Proposed Method

The proposed system operational diagram of presented in Figure 4.1. It outperforms underwater species classification using Convolutional Neural Networks (CNNs) and Support Vector Machines (SVMs).

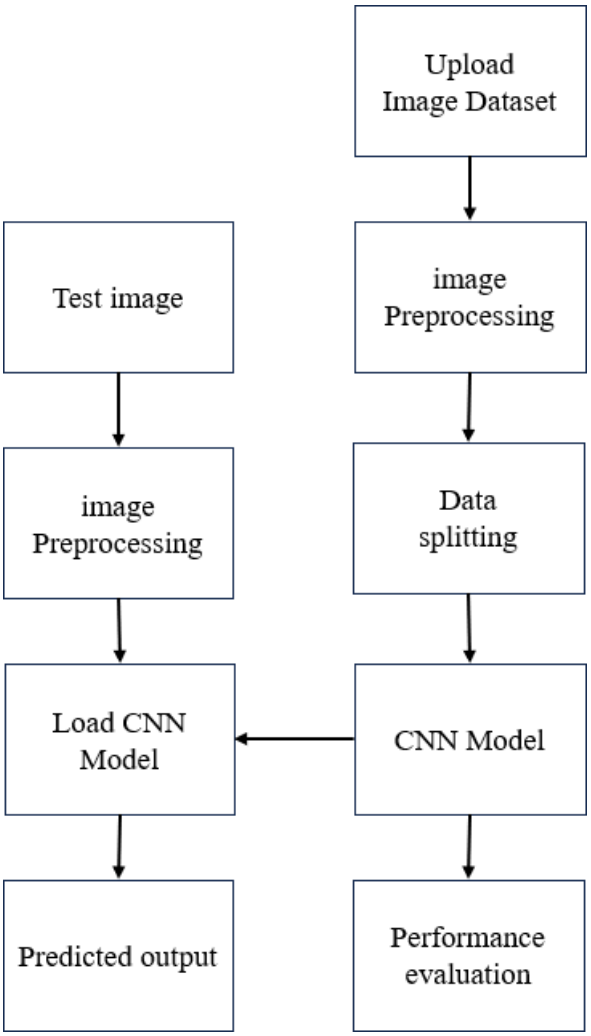


Figure 4.1: Block diagram of proposed model

- Importing Libraries: The necessary libraries are imported, including Tkinter for GUI, various machine learning libraries (e.g., Keras, scikit-learn), and computer vision libraries (e.g., OpenCV).
- Tkinter Setup: The main Tkinter window is created with the title "Underwater Species Classification using CNNs A Case Study on Sea Animals" and a specified geometry.

- Global Variables: Global variables like filename, X, Y, model, and accuracy are declared to store relevant information throughout the script.
- getID Function: A function getID(name) is defined, which assigns numerical labels to different categories (species) based on the category name.
- uploadDataset Function: The uploadDataset function is triggered by a button press, allowing the user to select a directory containing the dataset. It displays a message indicating that the dataset has been loaded.
- SVM Image Preprocessing Function: The svm Image Preprocessing function processes and loads image data for SVM training. It creates arrays (flat_data and target) and visualizes the distribution of categories using a count plot.
- Image Processing Function: The image Processing function displays statistics about the loaded dataset and provides options for further actions, such as building a CNN model.
- CNN Model Function: The cnn Model function builds a CNN model for species classification. It either loads a pre-trained model or trains a new one, saving the model architecture, weights, and training history. The model's accuracy is displayed.
- svmModel Function: The svmModel function builds an SVM model for species classification. It either loads a pre-trained model or trains a new one, saving the trained model. The SVM model's accuracy is displayed.
- predict Function: The predict function allows the user to upload a test image and make predictions using the trained model. The predicted species is displayed on the image.
- graph Function: The graph function displays a graph showing accuracy and loss over iterations during model training.
- close Function: The close function closes the main Tkinter window when the "Exit" button is pressed.
- Tkinter GUI Layout: Buttons for various actions (uploading dataset, preprocessing, building models, predicting, etc.) are created and placed on the

GUI window.A text widget is used to display information and messages to the user.

- Tkinter Main Loop: The main loop (main.mainloop()) keeps the Tkinter GUI window open, allowing the user to interact with the interface.

4.3 Image Preprocessing

Image preprocessing is a critical step in computer vision and image analysis tasks. It involves a series of operations to prepare raw images for further processing by algorithms or neural networks. Here's an explanation of each step in image preprocessing:

Step 0. Image Read: The first step in image preprocessing is reading the raw image from a source, typically a file on disk. Images can be in various formats, such as JPEG, PNG, BMP, or others. Image reading is performed using libraries or functions specific to the chosen programming environment or framework. The result of this step is a digital representation of the image that can be manipulated programmatically.

1. Image Resize: Image resize is a common preprocessing step, especially when working with machine learning models or deep neural networks. It involves changing the dimensions (width and height) of the image. Resizing can be necessary for several reasons:

- Ensuring uniform input size: Many machine learning models, especially convolutional neural networks (CNNs), require input images to have the same dimensions. Resizing allows you to standardize input sizes.
- Reducing computational complexity: Smaller images require fewer computations, which can be beneficial for faster training and inference.
- Managing memory constraints: In some cases, images need to be resized to fit within available memory constraints.

When resizing, it's essential to maintain the aspect ratio to prevent image distortion. Typically, libraries like OpenCV or Pillow provide convenient functions for resizing images.

2. Image to Array: In this step, the image is converted into a numerical representation in the form of a multidimensional array or tensor. Each pixel in the image corresponds

to a value in the array. The array is usually structured with dimensions representing height, width, and color channels (if applicable).

For grayscale images, the array is 2D, with each element representing the intensity of a pixel. For color images, it's a 3D or 4D array, with dimensions for height, width, color channels (e.g., Red, Green, Blue), and potentially batch size (if processing multiple images simultaneously).

The conversion from an image to an array allows for numerical manipulation and analysis, making it compatible with various data processing libraries and deep learning frameworks like NumPy or TensorFlow.

3. **Image to Float32:** Most machine learning and computer vision algorithms expect input data to be in a specific data type, often 32-bit floating-point numbers (float32). Converting the image array to float32 ensures that the pixel values can represent a wide range of intensities between 0.0 (black) and 1.0 (white) or sometimes between -1.0 and 1.0, depending on the specific normalization used.

This step is essential for maintaining consistency in data types and enabling compatibility with various machine learning frameworks and libraries. It's typically performed by dividing the pixel values by the maximum intensity value (e.g., 255 for an 8-bit image) to scale them to the [0.0, 1.0] range.

4. **Image to Binary:** Image binarization is a process of converting a grayscale image into a binary image, where each pixel is represented by either 0 (black) or 1 (white) based on a specified threshold. Binarization is commonly used for tasks like image segmentation, where you want to separate objects from the background.

The process involves setting a threshold value, and then for each pixel in the grayscale image, if the pixel value is greater than or equal to the threshold, it is set to 1; otherwise, it is set to 0.

Binarization simplifies the image and reduces it to essential information, which can be particularly useful in applications like character recognition or object tracking, where you need to isolate regions of interest.

4.4 Splitting the Dataset

In machine learning data preprocessing, we divide our dataset into a training set and test set. This is one of the crucial steps of data preprocessing as by doing this, we can enhance the performance of our machine learning model. Suppose if we have given training to our machine learning model by a dataset and we test it by a completely different dataset. Then, it will create difficulties for our model to understand the correlations between the models. If we train our model very well and its training accuracy is also very high, but we provide a new dataset to it, then it will decrease the performance. So we always try to make a machine learning model which performs well with the training set and also with the test dataset. Here, we can define these datasets as:

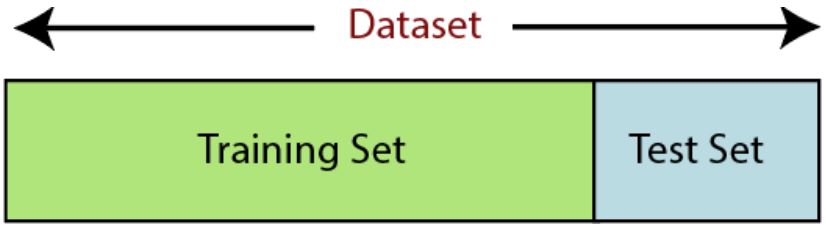


Figure 4.2: Splitting the dataset.

Training Set: A subset of dataset to train the machine learning model, and we already know the output.

Test set: A subset of dataset to test the machine learning model, and by using the test set, model predicts the output.

For splitting the dataset, we will use the below lines of code: from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test= train_test_split(x, y, test_size= 0.2, random_state=0)

Explanation

- In the above code, the first line is used for splitting arrays of the dataset into random train and test subsets.

- In the second line, we have used four variables for our output that are

- x_test: features for testing data

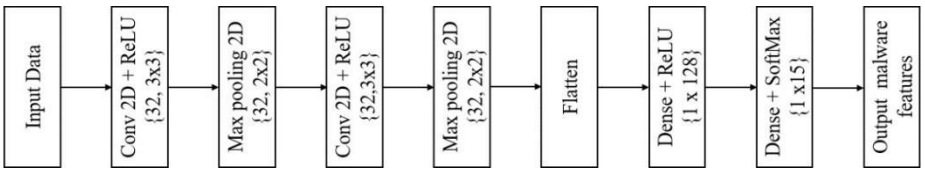
- y_train: Dependent variables for training data

- y_test: Independent variable for testing data

- In train_test_split() function, we have passed four parameters in which first two are for arrays of data, and test_size is for specifying the size of the test set. The test_size maybe .5, .3, or .2, which tells the dividing ratio of training and testing sets.
- The last parameter random_state is used to set a seed for a random generator so that you always get the same result, and the most used value for this is 42.

4.5 Convolutional Neural Network

Convolutional neural networks (CNNs) are a class of deep neural networks that are particularly effective for visual image analysis. It is successfully used in various applications such as image and video recognition, image segmentation, medical image analysis, and natural language processing. The CNN algorithm is designed to learn the features of the spatial structure automatically and adaptively from the input images.



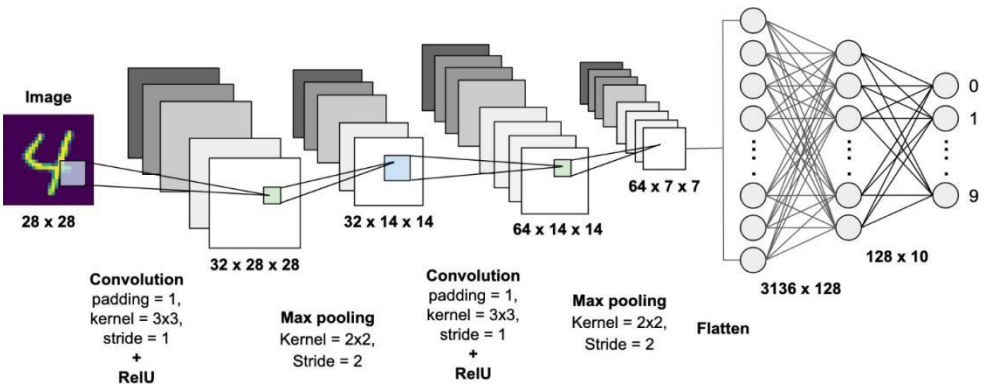


Figure 4.2. Proposed CNN Model.

4.5.1 Convolutional layer:

The Convolution Layer is the core constructing block of a Convolutional Neural Network (CNN), which plays a vital function inside the community's potential to method and analyze photographs. This layer is answerable for the majority of the computational efforts inside a CNN. It operates by using making use of numerous filters (additionally known as kernels) throughout the input image to provide what are called feature maps.

Each clear out within the convolutional layer is a small, but three-dimensional, array of numbers. "Small" refers to the spatial dimensions (top and width) being lots less than those of the input image, at the same time as "three-dimensional" way the filter extends through the entire depth of the enter quantity. For example, inside the case of an RGB photograph, which has a intensity of three (representing the purple, green, and blue channels), the filter out might also have a intensity of 3 to in shape the input.

The process of applying a filter involves sliding (or "convolving") the clear out throughout the peak and width of the input photograph and computing dot merchandise among the entries of the filter and the enter at each role. This operation extracts excessive-stage functions such as edges, corners, or textures from the enter photograph, relying on what the clear out has found out to stumble on for the duration of the training segment.

During training, through the backpropagation set of rules and optimization strategies like gradient descent, these filters are automatically adjusted to become aware of useful functions that assist in reaching the project at hand, together with classifying pix. The filters start off as random values however regularly adjust to detect particular types of features (like edges in various orientations, textures, or styles) which was important for classification or other duties.

4.5.2 Pooling Layer

The pooling layer is a crucial thing in Convolutional Neural Networks (CNNs) that usually follows the activation characteristic, which includes ReLU, inside the network structure. Its primary cause is to reduce the spatial length of the function maps produced via the convolutional layers even as maintaining the maximum essential statistics. This

discount in spatial length facilitates in controlling the number of parameters and computational load inside the network, thereby helping in stopping overfitting.

Pooling is done by way of dividing the characteristic map into disjoint regions (typically squares) and then applying a pooling operation inside every vicinity to achieve a unmarried output cost. The maximum not unusual pooling operation is max pooling, wherein the maximum value within every location is retained and the rest are discarded. However, different pooling strategies like common pooling exist, wherein rather than taking the maximum fee, the common value inside each location is computed.

Max pooling is popular because it facilitates in keeping the most distinguished capabilities detected by the convolutional layers, making the community extra robust to variations in enter and enhancing translation invariance (the capability to stumble on features no matter their location inside the enter). On the alternative hand, common pooling can now and again easy out the capabilities, which was useful in positive eventualities or architectures.

By reducing the spatial dimensions of the feature maps, pooling layers help in growing a more compact and summary representation of the enter data, that is beneficial for next layers to research higher-stage capabilities and styles. Additionally, pooling helps in lowering the computational cost of the network with the aid of down sampling the characteristic maps, that is specifically high-quality whilst coping with large input pics.

4.5.3 Fully Connected (FC) Layer

The Fully Connected (FC) layer (also known as Dense Layer) is a key factor that typically appears toward the give up of a Convolutional Neural Network (CNN) architecture, after the convolutional and pooling layers have processed the enter image. The primary function of FC layers is to perform high-degree reasoning and make decisions primarily based on the functions extracted and abstracted by means of the earlier layers within the community.

In a completely linked layer, every neuron is hooked up to every activation from the preceding layer. This way that the input to a completely connected layer is the flattened output of all previous layers, transformed into a single vector. This vector is then processed through one or more absolutely related layers, where every layer's output is calculated the use of matrix multiplication of the input vector with a weight matrix, observed via adding a bias term.

y=Wx+ b, where x is the input vector, W is the load matrix, b is the prejudice vector, and y is the output vector.

The absolutely related layers are crucial for the CNN as they permit the network to make feel of the features that have been identified and pooled by the convolutional and pooling layers. By having each neuron connected to all activations in the previous layer, the completely linked layer can examine complicated patterns and relationships within the information, allowing the community to categorise the input photograph into categories, apprehend gadgets, or perform other obligations relying at the final output layer and the loss characteristic used in the course of education. ss

Typically, the ultimate fully connected layer's output size corresponds to the variety of lessons in a category task, and a softmax activation feature is regularly carried out to this closing layer to produce a possibility distribution over the class.

4.5.4 ReLU Activation Layer

The Activation Layer is a fundamental aspect in Convolutional Neural Networks (CNNs) and other forms of neural networks, generally positioned proper after each convolutional layer. Its primary role is to introduce non-linearity into the network, that's essential for the network if you want to examine and model complicated styles inside the statistics.

One of the maximum usually used activation features in CNNs is the Rectified Linear Unit (ReLU). The ReLU characteristic is described mathematically as f(x)=max(zero,x), which means that it outputs the enter directly if it is effective, and outputs zero in any other case. This simple, piecewise linear characteristic is relatively powerful in increasing the nonlinear homes of the selection characteristic and of the overall network with out affecting the receptive fields of the convolution layer.

The inclusion of non-linearity is vital because with out it, the network, no matter how many layers it has, would essentially behave like a unmarried linear classifier, which limits its capability to research complex styles and perform duties including image type, item detection, etc. Non-linear activation features like ReLU assist the network in studying nonlinear mappings from inputs to outputs, enabling the modelling of complex information along with photos, speech, and textual content.

ReLU is especially famous due to its computational simplicity, which leads to quicker education, and its potential to mitigate the vanishing gradient trouble— a situation where gradients turn out to be too small for the community to analyse successfully— compared to conventional sigmoid or tanh activation capabilities. However, ReLU isn't without its issues, such as the "demise ReLU" trouble, in which neurons can every so often end up inactive and stop to make contributions to the getting to know process. Variants o f ReLU, such as Leaky ReLU and Parametric ReLU, have been proposed to address this and other limitations.

4.5.5 Flatten Layer

The Flatten layer in a Convolutional Neural Network (CNN) serves an easy but important function: it transforms its multidimensional enter right into a one-dimensional (1D) array. This process is important when transitioning from convolutional and pooling layers, which cope with information in more than one dimensions (width, top, depth), to fully connected (FC) layers, which require enter in a linear layout.

In CNN processing pixel, after several rounds of convolution and pooling, the community has learned a sequence of high-level functions represented in a multidimensional format. Before those features was used for type or regression obligations inside the fully related layers, they need to be "flattened" right into a vector

The Flatten layer does no longer contain any parameters or computation; it without a doubt reorganizes the information, making it well matched for the dense layers that observe. By converting the facts into a format suitable for category or regression, the Flatten layer acts as a bridge between the feature extraction and sample studying abilities of the convolutional layers and the selection-making capability of the completely connected layers.

4.5.6 Softmax Classifier

The softmax classifier is a mathematical characteristic typically used at the output layer of a neural community model, particularly in obligations related to classification wherein the objective is to assign the enter into considered one of several classes. It is particularly generic in fashions coping with multi-magnificence classification troubles.

The number one feature of the softmax classifier is to convert the uncooked output scores (often referred to as logits) from the neural community into chances by way of taking the exponential of every output and then normalizing these values by using dividing through the sum of all of the exponentials. This method ensures that the output values are inside the variety (zero,1) and sum up to at least one, making them interpretable as probabilities. The method for the softmax feature for a unmarried output unit i in a layer with K units is given by using:

Adam Optimizer: The Adam optimizer is a popular algorithm in the field of deep learning for optimizing neural network weights. It combines the best properties of the AdaGrad and RMSProp algorithms to provide an optimization scheme that can handle sparse gradients on noisy problems. Adam stands for Adaptive Moment Estimation and is known for its efficiency in terms of computational resource usage and for the little memory it requires.

Binary Cross entropy: Binary cross entropy, also known as log loss, is a common loss function used for binary classification problems in deep learning. In binary classification, the goal is to predict whether an input belongs to one of two classes (0 or 1). The logic behind binary cross entropy is that we want to penalize the model when its predicted probabilities diverge from the actual label. It measures the difference between two probability distributions - the predicted and the actual. Formally, for a single data point, the binary cross entropy loss is calculated as:

$$L(y, p) = -(y * log(p) + (1 - y) * log(1 - p))$$

Here, y is the true label (0 or 1), p is the predicted probability of the positive class

If the model predicts close to 0 when the actual label is 1, or predicts close to 1 when the actual label is 0, the loss will be high. The more divergent the predicted probability is from the actual label, the higher the loss. When training a CNN for binary classification, we calculate this log loss over the entire training set by taking the average:

$$Loss = -1/N * \Sigma (y * log(p) + (1 - y) * log(1 - p))$$

Where N is the number of training examples. Minimizing this loss function via gradient descent pushes the model to predict probabilities close to the actual labels and improve classification accuracy.

4.6 Advantages of CNN

- No human supervision
- High accuracy at image recognition
- Automatic feature extraction
- Weight sharing
- Ability to handle large datasets
- Minimize computation
- Uses Same knowledge across all image location

CHAPTER 5

UML DIAGRAMS

Class diagram: Class diagram is a static diagram. It represents the static view of an application.

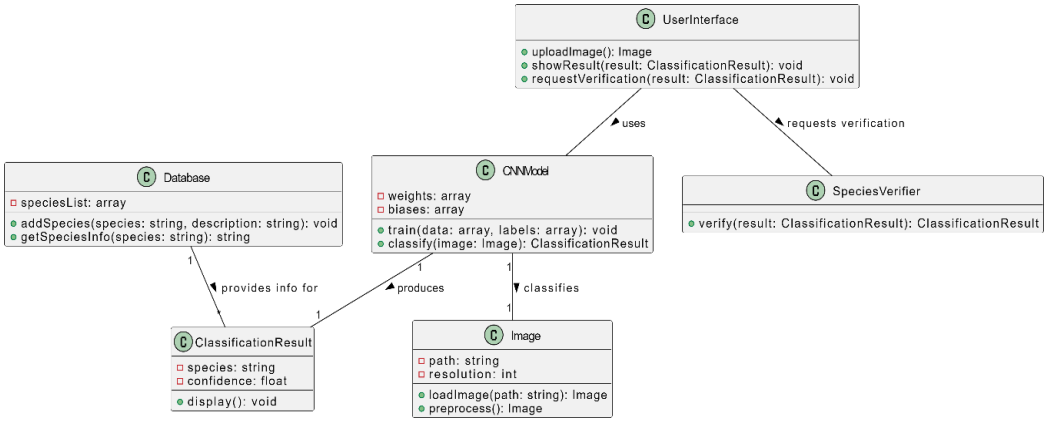


Fig . 5.1 : Class Diagram

Sequence Diagram: Sequence diagram is an interaction diagram that details how operations are carried out.

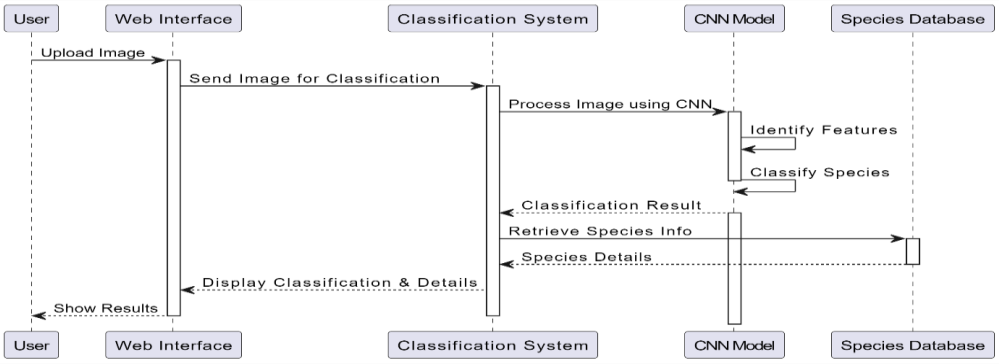


Fig. 5.2 : Sequence Diagram

Activity Diagram: Activity diagram is another important diagram in UML to

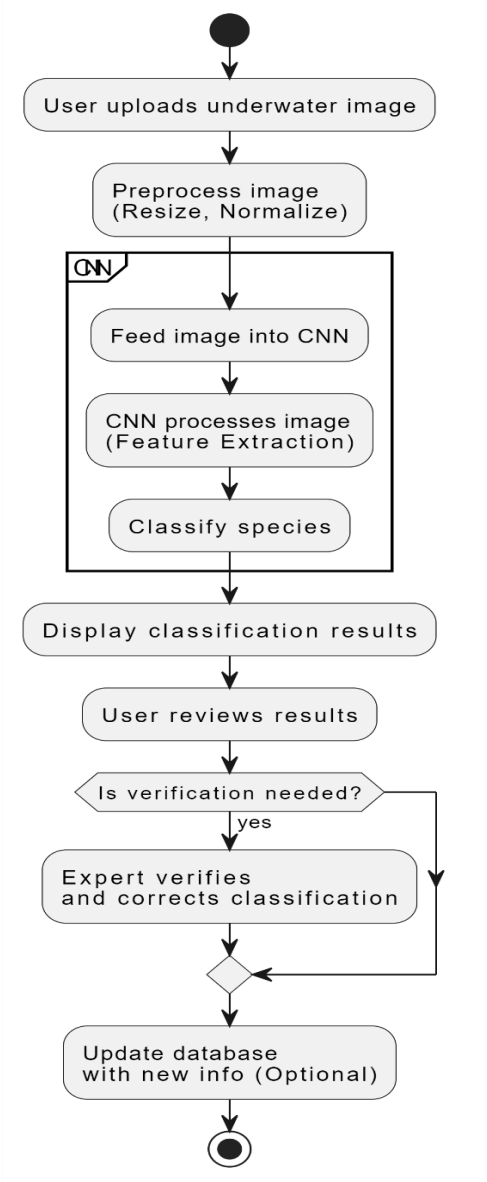


Fig. 5.3 : Activity Diagram

Deployment Diagram: Deployment diagram visualizes the physical hardware on which the software it will be deployed.

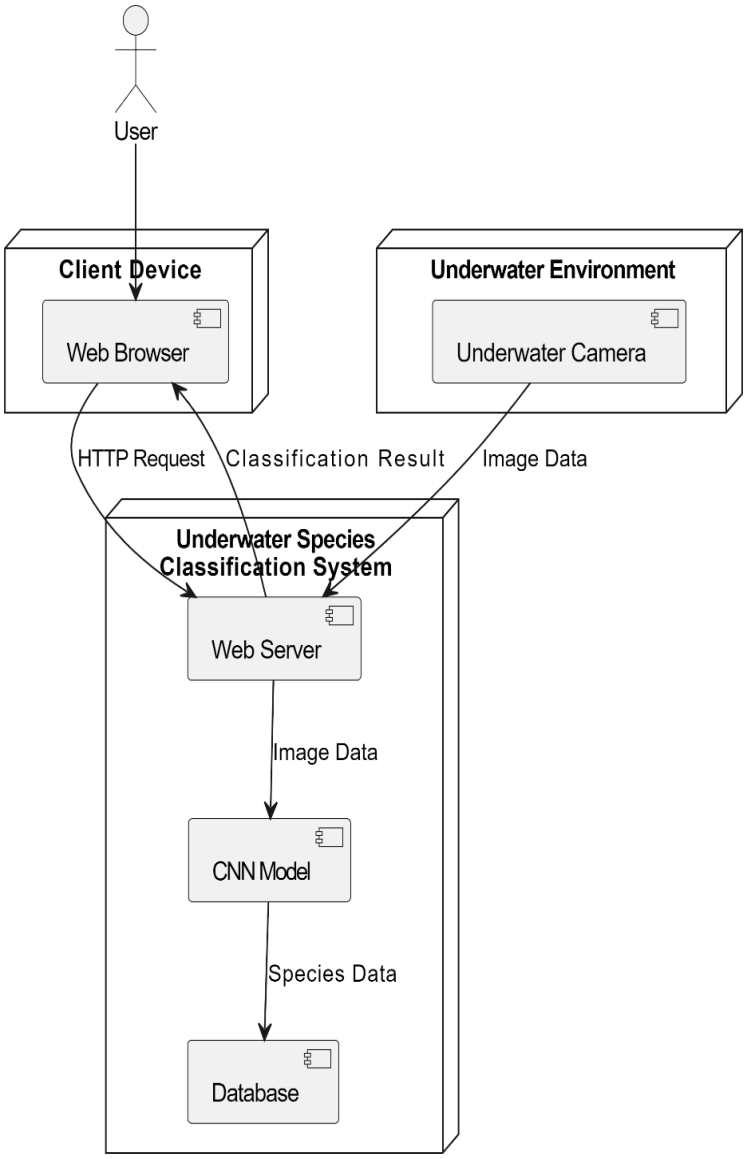


Fig. 5.4 : Deployment Diagram

<u>Use-case diagram</u>: The purpose of use case diagram is to capture the dynamic aspects of a system.

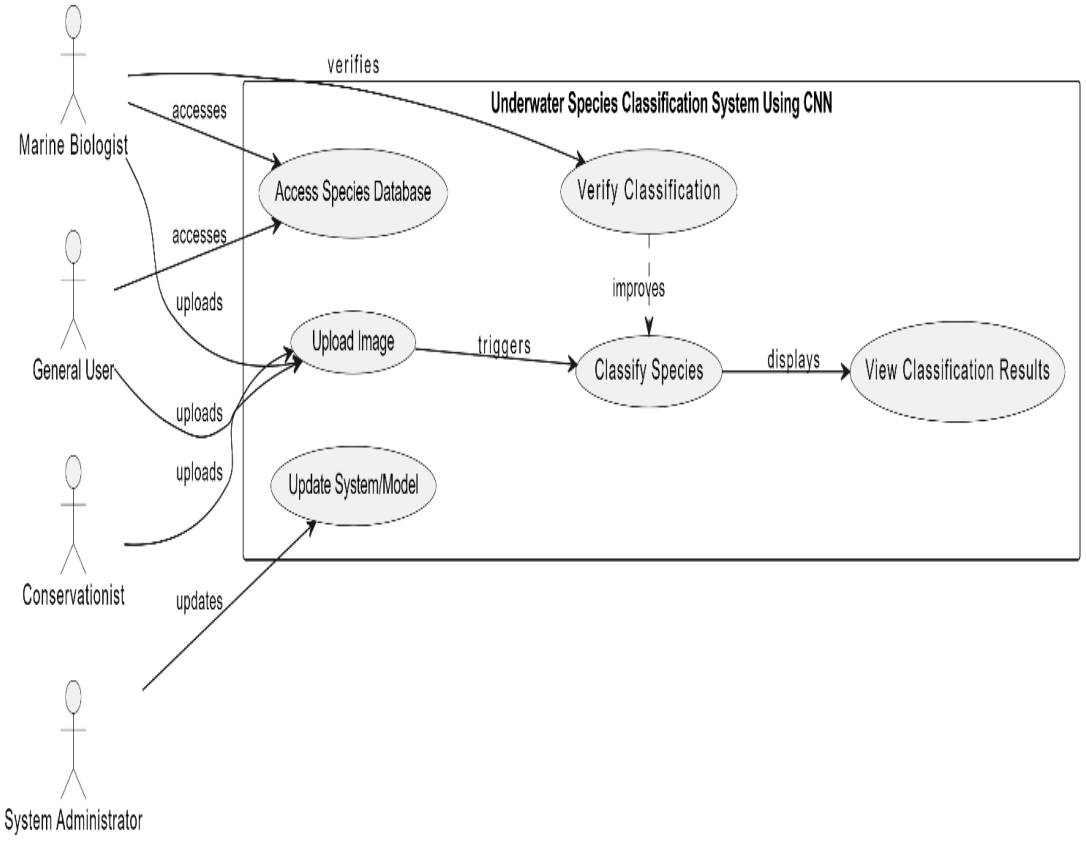


Fig. 5.5 : Use-Case Diagram

Component Diagram: Component diagram describes the organization and wiring of the physical components in a system.

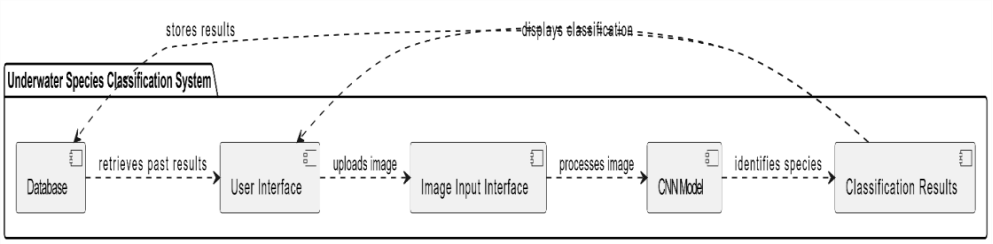


Fig. 5.6 : Component Diagram

Data-Flow Diagram: Data Flow Diagram (DFD) is a graphical or visual representation using a standardized set of symbols and notations to describe a business's operations through data movement.

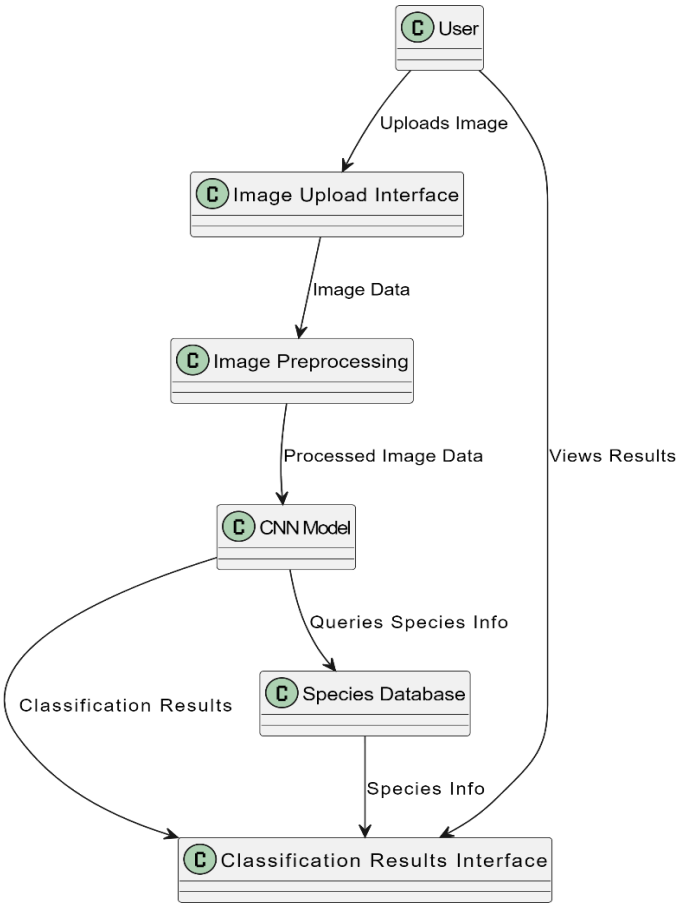


Fig. 5.7 : Data-Flow Diagram

CHAPTER 6

SOFTWARE ENVIRONMENT

What is Python?

Below are some facts about Python.

- Python is currently the most widely used multi-purpose, high-level programming language.
- Python allows programming in Object-Oriented and Procedural paradigms. Python programs generally are smaller than other programming languages like Java.
- Programmers have to type relatively less and indentation requirement of the language, makes them readable all the time.
- Python language is being used by almost all tech-giant companies like – Google, Amazon, Facebook, Instagram, Dropbox, Uber… etc.

The biggest strength of Python is huge collection of standard library which was used for the following –

- Machine Learning
- GUI Applications (like Kivy, Tkinter, PyQt etc.)
- Web frameworks like Django (used by YouTube, Instagram, Dropbox)
- Image processing (like Opencv, Pillow)
- Web scraping (like Scrapy, BeautifulSoup, Selenium)
- Test frameworks
- Multimedia

Advantages of Python

Let's see how Python dominates over other languages.

1. **Extensive Libraries**

Python downloads with an extensive library and it contain code for various purposes like regular expressions, documentation-generation, unit-testing, web browsers,

threading, databases, CGI, email, image manipulation, and more. So, we don't have to write the complete code for that manually.

2. **Extensible**

   As we have seen earlier, Python was extended to other languages. You can write some of your code in languages like C++ or C. This comes in handy, especially in projects.

3. **Embeddable**

   Complimentary to extensibility, Python is embeddable as well. You can put your Python code in your source code of a different language, like C++. This lets us add scripting capabilities to our code in the other language.

4. **Improved Productivity**

   The language's simplicity and extensive libraries render programmers more productive than languages like Java and C++ do. Also, the fact that you need to write less and get more things done.

5. **IOT Opportunities**

   Since Python forms the basis of new platforms like Raspberry Pi, it finds the future bright for the Internet Of Things. This is a way to connect the language with the real world.

6. **Simple and Easy**

   When working with Java, you may have to create a class to print 'Hello World'. But in Python, just a print statement will do. It is also quite easy to learn, understand, and code. This is why when people pick up Python, they have a hard time adjusting to other more verbose languages like Java.

7. **Readable**

   Because it is not such a verbose language, reading Python is much like reading English. This is the reason why it is so easy to learn, understand, and code. It also does not need curly braces to define blocks, and indentation is mandatory. This further aids the readability of the code.

8. **Object-Oriented**

This language supports both the procedural and object-oriented programming paradigms. While functions help us with code reusability, classes and objects let us model the real world. A class allows the encapsulation of data and functions into one.

9. **Free and Open-Source**

   Like we said earlier, Python is freely available. But not only can you download Python for free, but you can also download its source code, make changes to it, and even distribute it. It downloads with an extensive collection of libraries to help you with your tasks.

10. **Portable**

    When you code your project in a language like C++, you may need to make some changes to it if you want to run it on another platform. But it isn't the same with Python. Here, you need to code only once, and you can run it anywhere. This is called Write Once Run Anywhere (WORA). However, you need to be careful enough not to include any system-dependent features.

11. **Interpreted**

    Lastly, we will say that it is an interpreted language. Since statements are executed one by one, debugging is easier than in compiled languages.

    Any doubts till now in the advantages of Python? Mention in the comment section.

Advantages of Python Over Other Languages

1. **Less Coding**

   Almost all of the tasks done in Python requires less coding when the same task is done in other languages. Python also has an awesome standard library support, so you don't have to search for any third-party libraries to get your job done. This is the reason that many people suggest learning Python to beginners.

2. **Affordable**

   Python is free therefore individuals, small companies or big organizations can leverage the free available resources to build applications. Python is popular and widely used so it gives you better community support.

The 2019 Github annual survey showed us that Python has overtaken Java in the most popular programming language category.

3. **Python is for Everyone**

   Python code can run on any machine whether it is Linux, Mac or Windows. Programmers need to learn different languages for different jobs but with Python, you can professionally build web apps, perform data analysis and machine learning, automate things, do web scraping and also build games and powerful visualizations. It is an all-rounder programming language.

Disadvantages of Python

So far, we've seen why Python is a great choice for your project. But if you choose it, you should be aware of its consequences as well. Let's now see the downsides of choosing Python over another language.

1. **Speed Limitations**

   We have seen that Python code is executed line by line. But since Python is interpreted, it often results in slow execution. This, however, isn't a problem unless speed is a focal point for the project. In other words, unless high speed is a requirement, the benefits offered by Python are enough to distract us from its speed limitations.

2. **Weak in Mobile Computing and Browsers**

   While it serves as an excellent server-side language, Python is much rarely seen on the client-side. Besides that, it is rarely ever used to implement smartphone-based applications. One such application is called Carbonnelle.

   The reason it is not so famous despite the existence of Brython is that it isn't that secure.

3. **Design Restrictions**

   As you know, Python is dynamically typed. This means that you don't need to declare the type of variable while writing the code. It uses duck-typing. But wait, what's that? Well, it just means that if it looks like a duck, it must be a duck. While this is easy on the programmers during coding, it can raise run-time errors.

4. **Underdeveloped Database Access Layers**

   Compared to more widely used technologies like JDBC (Java DataBase Connectivity) and ODBC (Open DataBase Connectivity), Python's database access layers are a bit underdeveloped. Consequently, it is less often applied in huge enterprises.

5. **Simple**

   No, we're not kidding. Python's simplicity can indeed be a problem. Take my example. I don't do Java, I'm more of a Python person. To me, its syntax is so simple that the verbosity of Java code seems unnecessary.

   This was all about the Advantages and Disadvantages of Python Programming Language.

Modules Used in Project

NumPy

NumPy is a general-purpose array-processing package. It provides a high-performance multidimensional array object, and tools for working with these arrays.

It is the fundamental package for scientific computing with Python. It contains various features including these important ones:

- A powerful N-dimensional array object
- Sophisticated (broadcasting) functions
- Tools for integrating C/C++ and Fortran code
- Useful linear algebra, Fourier transform, and random number capabilities

Besides its obvious scientific uses, NumPy can also be used as an efficient multi-dimensional container of generic data. Arbitrary datatypes was defined using NumPy which allows NumPy to seamlessly and speedily integrate with a wide variety of databases.

Pandas

Pandas is an open-source Python Library providing high-performance data manipulation and analysis tool using its powerful data structures. Python was majorly used for data munging and preparation. It had very little contribution towards data analysis. Pandas solved this problem. Using Pandas, we can accomplish five typical

steps in the processing and analysis of data, regardless of the origin of data load, prepare, manipulate, model, and analyze. Python with Pandas is used in a wide range of fields including academic and commercial domains including finance, economics, Statistics, analytics, etc.

Matplotlib

Matplotlib is a Python 2D plotting library which produces publication quality figures in a variety of hardcopy formats and interactive environments across platforms. Matplotlib was used in Python scripts, the Python and Ipython shells, the Jupyter Notebook, web application servers, and four graphical user interface toolkits. Matplotlib tries to make easy things easy and hard things possible. You can generate plots, histograms, power spectra, bar charts, error charts, scatter plots, etc., with just a few lines of code. For examples, see the sample plots and thumbnail gallery.

For simple plotting the pyplot module provides a MATLAB-like interface, particularly when combined with Ipython. For the power user, you have full control of line styles, font properties, axes properties, etc, via an object oriented interface or via a set of functions familiar to MATLAB users.

Scikit – learn

Scikit-learn provides a range of supervised and unsupervised learning algorithms via a consistent interface in Python. It is licensed under a permissive simplified BSD license and is distributed under many Linux distributions, encouraging academic and commercial use. Python

Install Python Step-by-Step in Windows and Mac

Python a versatile programming language doesn’t come pre-installed on your computer devices. Python was first released in the year 1991 and until today it is a very popular high-level programming language. Its style philosophy emphasizes code readability with its notable use of great whitespace.

The object-oriented approach and language construct provided by Python enables programmers to write both clear and logical code for projects. This software does not come pre-packaged with Windows.

How to Install Python on Windows and Mac

There have been several updates in the Python version over the years. The question is how to install Python? It was confusing for the beginner who is willing to start learning Python but this tutorial will solve your query. The latest or the newest version of Python is version 3.7.4 or in other words, it is Python 3.

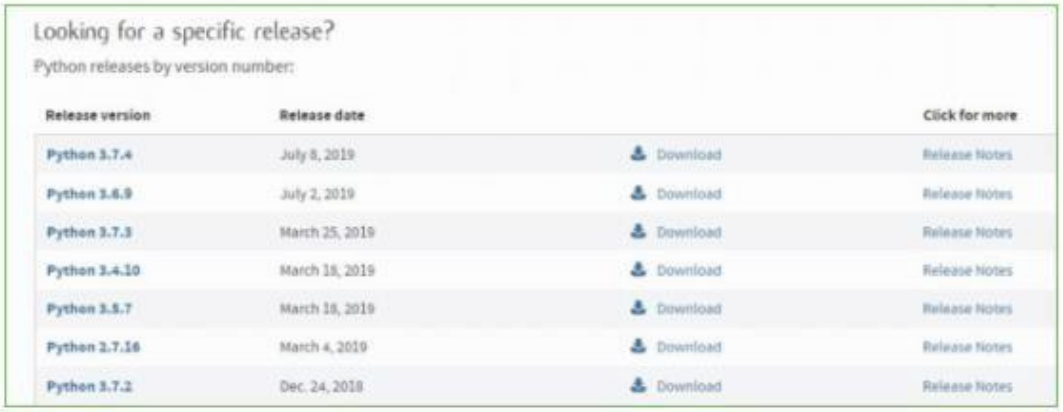
Note: The python version 3.7.4 cannot be used on Windows XP or earlier devices. Before you start with the installation process of Python. First, you need to know about your System Requirements. Based on your system type i.e. operating system and based processor, you must download the python version. My system type is a Windows 64- bit operating system. So the steps below are to install python version 3.7.4 on Windows 7 device or to install Python 3. Download the Python Cheatsheet here.The steps on how to install Python on Windows 10, 8 and 7 are divided into 4 parts to help understand better.

Download the Correct version into the system

Step 1: Go to the official site to download and install python using Google Chrome or any other web browser. OR Click on the following link: https://www.python.org

Now, check for the latest and the correct version for your operating system. Step 2: Click on the Download Tab.

Step 3: You can either select the Download Python for windows 3.7.4 button in Yellow Color or you can scroll further down and click on download with respective to their version. Here, we are downloading the most recent python version for windows 3.7.4



Looking for a specific release?

Python releases by version number:

Release version	Release date		Click for more
Python 3.7.4	July 8, 2019	Download	Release Notes
Python 3.6.9	July 2, 2019	Download	Release Notes
Python 3.7.3	March 25, 2019	Download	Release Notes
Python 3.4.10	March 18, 2019	Download	Release Notes
Python 3.5.7	March 18, 2019	Download	Release Notes
Python 2.7.16	March 4, 2019	Download	Release Notes
Python 3.7.2	Dec. 24, 2018	Download	Release Notes

Step 4: Scroll down the page until you find the Files option.

Step 5: Here you see a different version of python along with the operating system.

Files

Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		6811167e5b2db4aef7b9ab01bf0f9be	23017663	SIG
XZ compressed source tarball	Source release		d33e4aae66097051c1ecae5ee3604803	17131432	SIG
macOS 64-bit/32-bit installer	Mac OS X	for Mac OS X 10.6 and later	6428b4fa7583daff1a442cbabcee08e6	34898416	SIG
macOS 64-bit installer	Mac OS X	for OS X 10.9 and later	5dd605c38217a45773bf5e4a936b241f	28982845	SIG
Windows help file	Windows		d639995713a2c06b2ac5bcade6b4f7cd2	8131761	SIG
Windows x86-64 embeddable zip file	Windows	for AMD64/EM64T/x64	9b00c3cf6d9ee3b6a8e82184a4572fa2	7504391	SIG
Windows x86-64 executable installer	Windows	for AMD64/EM64T/x64	a702b4b0ad76de6cdc304c3a583e563400	26480368	SIG
Windows x86-64 web-based installer	Windows	for AMD64/EM64T/x64	28cb1c608f8bd73ae8e53a3bd351b4bd2	1362904	SIG
Windows x86 embeddable zip file	Windows		9fab38d1f8941879fda94133574139d8	6741626	SIG
Windows x86 executable installer	Windows		33cc802942a54446a3d845147639478f	25663848	SIG
Windows x86 web-based installer	Windows		1b670cfa6d317df82c3093ea371d87c	1324608	SIG

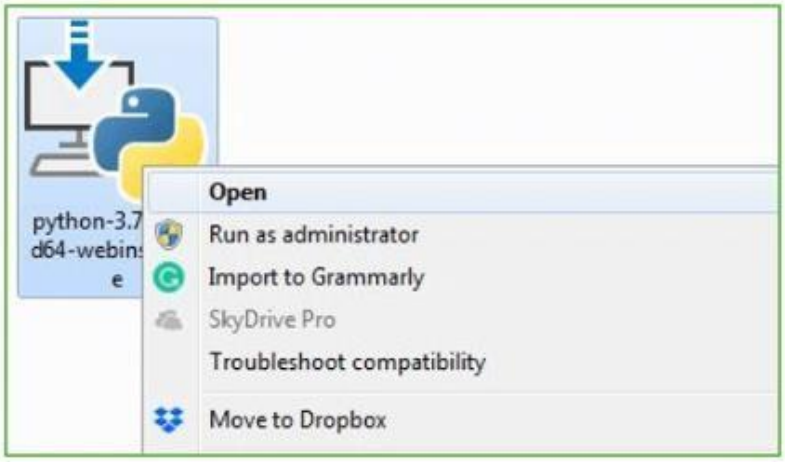
- To download Windows 32-bit python, you can select any one from the three options: Windows x86 embeddable zip file, Windows x86 executable installer or Windows x86 web-based installer.
- To download Windows 64-bit python, you can select any one from the three options: Windows x86-64 embeddable zip file, Windows x86-64 executable installer or Windows x86-64 web-based installer.

  Here we will install Windows x86-64 web-based installer. Here your first part regarding which version of python is to be downloaded is completed. Now we move ahead with the second part in installing python i.e. Installation

  Note: To know the changes or updates that are made in the version you can click on the Release Note Option.

Installation of Python

Step 1: Go to Download and Open the downloaded python version to carry out the installation process.



Step 2: Before you click on Install Now, Make sure to put a tick on Add Python 3.7 to PATH.

Step 3: Click on Install NOW After the installation is successful. Click on Close.

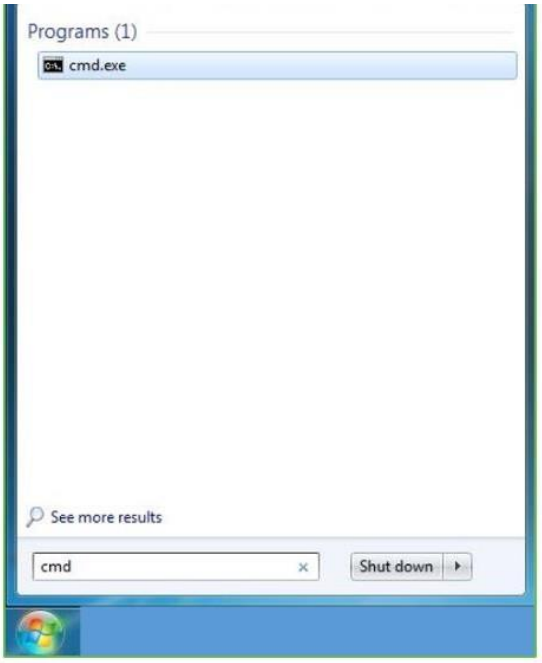
With these above three steps on python installation, you have successfully and correctly installed Python. Now is the time to verify the installation.

Note: The installation process might take a couple of minutes.

Verify the Python Installation

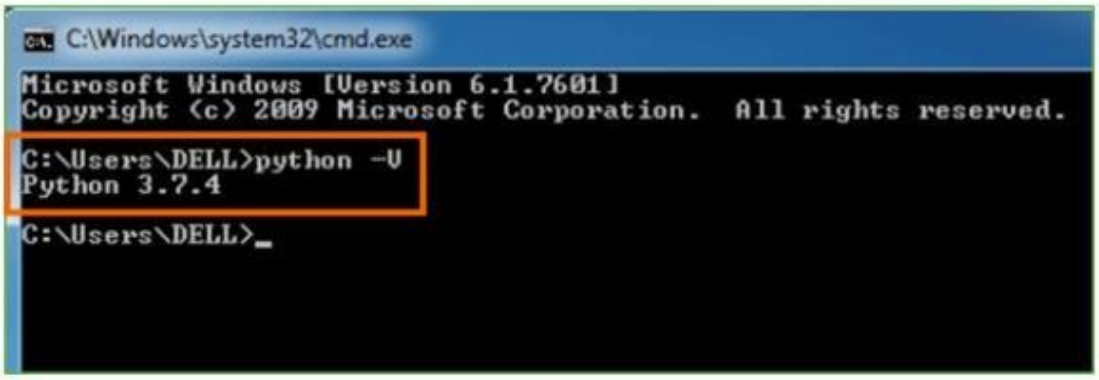
Step 1: Click on Start

Step 2: In the Windows Run Command, type “cmd”.



Step 3: Open the Command prompt option.

Step 4: Let us test whether the python is correctly installed. Type python –V and press Enter.



Step 5: You will get the answer as 3.7.4

Note: If you have any of the earlier versions of Python already installed. You must first uninstall the earlier version and then install the new one.

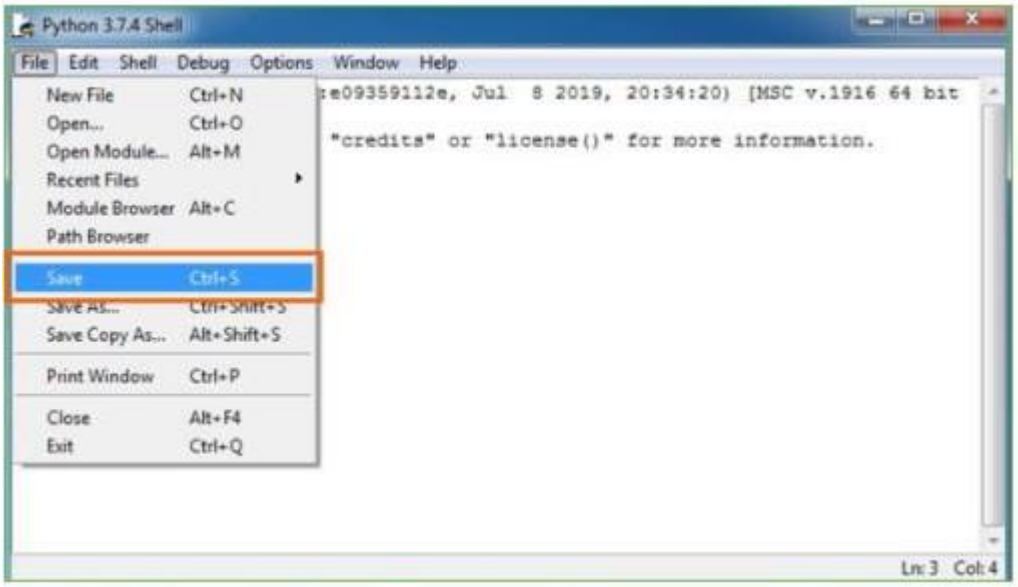
Check how the Python IDLE works

Step 1: Click on Start

Step 2: In the Windows Run command, type “python idle”.

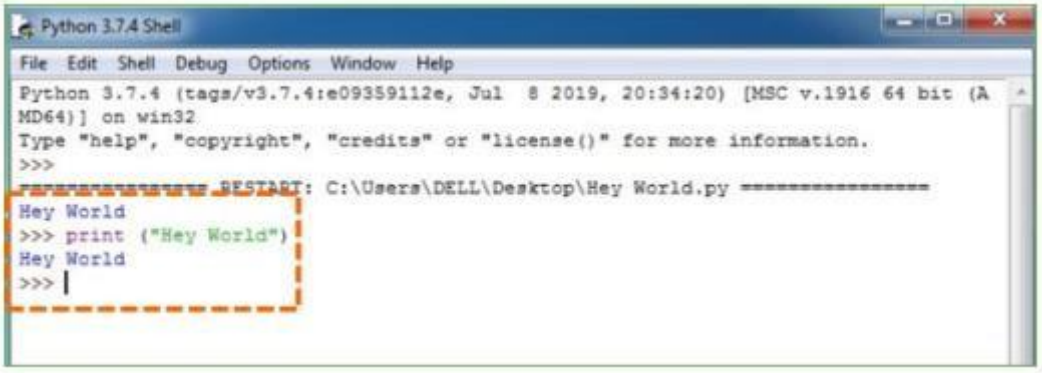
Step 3: Click on IDLE (Python 3.7 64-bit) and launch the program

Step 4: To go ahead with working in IDLE you must first save the file. Click on File > Click on Save



Step 5: Name the file and save as type should be Python files. Click on SAVE. Here I have named the files as Hey World.

Step 6: Now for e.g. enter print (“Hey World”) and Press Enter.



You will see that the command given is launched. With this, we end our tutorial on how to install Python. You have learned how to download python for windows into your respective operating system.

Note: Unlike Java, Python does not need semicolons at the end of the statements otherwise it won’t work.

CHAPTER 7

SYSTEM REQUIREMENTS

Software Requirements

The functional requirements or the overall description documents include the product perspective and features, operating system and operating environment, graphics requirements, design constraints and user documentation.

The appropriation of requirements and implementation constraints gives the general overview of the project in regard to what the areas of strength and deficit are and how to tackle them.

- Python IDLE 3.7 version (or)
- Anaconda 3.7 (or)
- Jupyter (or)
- Google colab

Hardware Requirements

Minimum hardware requirements are very dependent on the particular software being developed by a given Enthought Python / Canopy / VS Code user. Applications that need to store large arrays/objects in memory will require more RAM, whereas applications that need to perform numerous calculations or tasks more quickly will require a faster processor.

- Operating system : Windows, Linux
- Processor : minimum intel i3
- Ram : minimum 4 GB
- Hard disk : minimum 250GB

CHAPTER 8

FUNCTIONAL REQUIREMENTS

Output Design

Outputs from computer systems are required primarily to communicate the results of processing to users. They are also used to provides a permanent copy of the results for later consultation. The various types of outputs in general are:

- External Outputs, whose destination is outside the organization
- Internal Outputs whose destination is within organization and they are the
- User's main interface with the computer.
- Operational outputs whose use is purely within the computer department.
- Interface outputs, which involve the user in communicating directly.

Output Definition

The outputs should be defined in terms of the following points:

- Type of the output
- Content of the output
- Format of the output
- Location of the output
- Frequency of the output
- Volume of the output
- Sequence of the output

It is not always desirable to print or display data as it is held on a computer. It should be decided as which form of the output is the most suitable.

Input Design

Input design is a part of overall system design. The main objective during the input design is as given below:

- To produce a cost-effective method of input.

- To achieve the highest possible level of accuracy.
- To ensure that the input is acceptable and understood by the user.

Input Stages

The main input stages was listed as below:

- Data recording
- Data transcription
- Data conversion
- Data verification
- Data control
- Data transmission
- Data validation
- Data correction

Input Types

It is necessary to determine the various types of inputs. Inputs was categorized as follows:

- External inputs, which are prime inputs for the system.
- Internal inputs, which are user communications with the system.
- Operational, which are computer department's communications to the system?
- Interactive, which are inputs entered during a dialogue.

Input Media

At this stage choice has to be made about the input media. To conclude about the input media consideration has to be given to;

- Type of input
- Flexibility of format
- Speed
- Accuracy

- Verification methods
- Rejection rates
- Ease of correction
- Storage and handling requirements
- Security
- Easy to use
- Portability

Keeping in view the above description of the input types and input media, it was said that most of the inputs are of the form of internal and interactive. As Input data is to be the directly keyed in by the user, the keyboard was considered to be the most suitable input device.

Error Avoidance

At this stage care is to be taken to ensure that input data remains accurate form the stage at which it is recorded up to the stage in which the data is accepted by the system. This was achieved only by means of careful control each time the data is handled.

Error Detection

Even though every effort is make to avoid the occurrence of errors, still a small proportion of errors is always likely to occur, these types of errors was discovered by using validations to check the input data.

Data Validation

Procedures are designed to detect errors in data at a lower level of detail. Data validations have been included in the system in almost every area where there is a possibility for the user to commit errors. The system will not accept invalid data. Whenever an invalid data is keyed in, the system immediately prompts the user and the user has to again key in the data and the system will accept the data only if the data is correct. Validations have been included where necessary.

The system is designed to be a user friendly one. In other words the system has been designed to communicate effectively with the user. The system has been designed with popup menus.

User Interface Design

It is essential to consult the system users and discuss their needs while designing the user interface:

User Interface Systems Was Broadly Clasified As:

- User initiated interface the user is in charge, controlling the progress of the user/computer dialogue. In the computer-initiated interface, the computer selects the next stage in the interaction.
- Computer initiated interfaces

  In the computer-initiated interfaces the computer guides the progress of the user/computer dialogue. Information is displayed and the user response of the computer takes action or displays further information.

User Initiated Interfaces

User initiated interfaces fall into two approximate classes:

- Command driven interfaces: In this type of interface the user inputs commands or queries which are interpreted by the computer.
- Forms oriented interface: The user calls up an image of the form to his/her screen and fills in the form. The forms-oriented interface is chosen because it is the best choice.

Computer-Initiated Interfaces

The following computer – initiated interfaces were used:

- The menu system for the user is presented with a list of alternatives and the user chooses one; of alternatives.
- Questions – answer type dialog system where the computer asks question and takes action based on the basis of the users reply.

  Right from the start the system is going to be menu driven, the opening menu displays the available options. Choosing one option gives another popup menu with more options. In this way every option leads the users to data entry form where the user can key in the data.

Error Message Design

The design of error messages is an important part of the user interface design. As user is bound to commit some errors or other while designing a system the system should be designed to be helpful by providing the user with information regarding the error he/she has committed.

This application must be able to produce output at different modules for different inputs.

Performance Requirements

Performance is measured in terms of the output provided by the application. Requirement specification plays an important part in the analysis of a system. Only when the requirement specifications are properly given, it is possible to design a system, which will fit into required environment. It rests largely in the part of the users of the existing system to give the requirement specifications because they are the people who finally use the system. This is because the requirements have to be known during the initial stages so that the system was designed according to those requirements. It is very difficult to change the system once it has been designed and on the other hand designing a system, which does not cater to the requirements of the user, is of no use.

The requirement specification for any system was broadly stated as given below:

- The system should be able to interface with the existing system
- The system should be accurate
- The system should be better than the existing system
- The existing system is completely dependent on the user to perform all the duties.

CHAPTER 9

SOURCE CODE

```
from tkinter import messagebox from tkinter import *
from tkinter import simpledialog import tkinter
import warnings warnings.filterwarnings('ignore') import matplotlib.pyplot as plt
import numpy as np
from tkinter import ttk
from tkinter import filedialog
from keras.utils.np_utils import to_categorical from keras.models import
Sequential
from keras.layers.core import Dense,Activation,Dropout, Flatten from
sklearn.metrics import accuracy_score
import os
from sklearn.svm import SVC import cv2
from keras.layers import Convolution2D from keras.layers import MaxPooling2D
import pickle
from sklearn.model_selection import train_test_split from keras.models import
model_from_json
import pickle import pandas as pd
import numpy as np
import matplotlib.pyplot as plt import seaborn as sns
from keras.models import model_from_json from skimage.transform import
resize
from skimage.io import imread from skimage import io, transform


main = Tk()
main.title("Underwater Species Classification using CNNs A Case Study on Sea
Animals")
main.geometry("1300x1200")



global filename global X, Y global model global accuracy
```

```
    animals = ['Jelly Fish','Dolphin','Octopus'] def getID(name):
        # Replace this with your implementation to assign labels based on the
    directory name
    # For example:
if name == "Jelly Fish": return 0
elif name == "Dolphin": return 1
elif name == "Octopus": return 2
    else:
    return -1  # Return an appropriate label for other cases



def uploadDataset(): global X, Y global filename
    text.delete('1.0', END)
    filename = filedialog.askdirectory(initialdir=".") text.insert(END,'dataset
    loaded\n')

    def svmImagePreprocessing():
    global x_test,x_train,y_test,y_train,df Categories=['Jelly Fish','Dolphin']

    flat_data_arr=[] #input array target_arr=[] #output array datadir=r"data"
        #create file paths by combining the datadir (data directory) with the filenames
    'flat_data.npy
    flat_data_file = os.path.join(datadir, 'flat_data.npy') target_file =
    os.path.join(datadir, 'target.npy')
if os.path.exists(flat_data_file) and os.path.exists(target_file): # Load the existing
    arrays
    flat_data = np.load(flat_data_file)
    target = np.load(target_file) #dataframe df=pd.DataFrame(flat_data)
    df['Target']=target



    x=df.iloc[:,:-1]
    y=df.iloc[:,-1]
```

```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.40,random_state=77)


        text.insert(END,"Total number of images found in dataset is  :
"+str(np.shape(x))+"\n")
text.insert(END,"Total Diseases found in dataset is : "+str(np.shape(y))+"\n")


plt.figure(figsize=(8, 6))
ax = sns.countplot(data=df, x='Target') plt.xlabel('Target', fontsize=12)
plt.ylabel('Count', fontsize=12) plt.title('Count Plot for Target', fontsize=14) #
Add count labels on top of the bars
for p in ax.patches: ax.annotate(f"{p.get_height()}",
(p.get_x() + p.get_width() / 2., p.get_height()),
ha='center', va='center', fontsize=11, color='black', xytext=(0, 5),
textcoords='offset points')
plt.show()


else:
#path which contains all the categories of images for i in Categories:
print(f'loading... category : {i}') path=os.path.join(datadir,i)
#create file paths by combining the datadir (data directory) with the i for img in
os.listdir(path):
img_array=imread(os.path.join(path,img))#Reads the image using imread.
        img_resized=resize(img_array,(150,150,3)) #Resizes the image to a
common size of (150, 150, 3) pixels.
        flat_data_arr.append(img_resized.flatten()) #Flattens the resized image
array and adds it to the flat_data_arr.
        target_arr.append(Categories.index(i)) #Adds the index of the category
to the target_arr.
print(f'loaded category:{i} successfully')
```

```
                #After processing all images, it converts the lists to NumPy arrays
    (flat_data and target).
    flat_data=np.array(flat_data_arr) target=np.array(target_arr)
    # Save the arrays(flat_data ,target ) into the files(flat_data.npy,target.npy)
    np.save(os.path.join(datadir, 'flat_data.npy'), flat_data)
    np.save(os.path.join(datadir, 'target.npy'), target)



    def imageProcessing():
    text.delete('1.0', END) global X, Y
    #global speech_X, speech_Y


    X = np.load('model/X.txt.npy') Y = np.load('model/Y.txt.npy')
    text.insert(END,"Total number of images found in dataset is  :
    "+str(len(X))+"\n") text.insert(END,"Total Species found in dataset is :
    "+str(animals)+"\n")

def cnnModel(): global model global accuracy
    text.delete('1.0', END)
    if os.path.exists('model/model.json'):
with open('model/model.json', "r") as json_file: loaded_model_json =
    json_file.read()
        model = model_from_json(loaded_model_json) json_file.close()
    model.load_weights("model/model_weights.h5")
    model._make_predict_function() print(model.summary())
    f = open('model/history.pckl', 'rb') accuracy = pickle.load(f)
    f.close()
    acc = accuracy['accuracy'] acc = acc[9] * 100
        text.insert(END,"CNN  Model Accuracy = "+str(acc)) else:
    model = Sequential() #resnet transfer learning code here
            model.add(Convolution2D(32, 3, 3, input_shape = (64, 64, 3), activation =
    'relu'))
```

```
model.add(MaxPooling2D(pool_size = (2, 2)))
model.add(Convolution2D(32, 3, 3, activation = 'relu'))
model.add(MaxPooling2D(pool_size = (2, 2))) model.add(Flatten())
model.add(Dense(output_dim = 256, activation = 'relu'))
model.add(Dense(output_dim = 3, activation = 'softmax'))
model.compile(optimizer = 'adam', loss = 'categorical_crossentropy', metrics =
['accuracy'])
print(model.summary())
        hist = model.fit(X, Y, batch_size=16, epochs=10, validation_split=0.2,
shuffle=True, verbose=2)
model.save_weights('model/model_weights.h5') model_json = model.to_json()
with open("model/model.json", "w") as json_file: json_file.write(model_json)
json_file.close()
f = open('model/history.pckl', 'wb') pickle.dump(hist.history, f) f.close()
f = open('model/history.pckl', 'rb') accuracy = pickle.load(f)
f.close()
acc = accuracy['accuracy'] acc = acc[9] * 100
text.insert(END,"CNN  Model Prediction Accuracy = "+str(acc))


def svmModel():
global model,x_train,y_train global accuracy text.delete('1.0', END)
if os.path.exists('model/SVM_Classifier'):
with open('model/SVM_Classifier', 'rb') as model_file: model =
pickle.load(model_file) y_pred=model.predict(x_test)
acc=accuracy_score(y_test,y_pred) text.insert(END,"SVM Model Accuracy =
"+str(acc))
else:
model = SVC(kernel='linear', C=1.0, random_state=42) model.fit(x_train,y_train)
y_pred=model.predict(x_test) Acc=accuracy_score(y_test,y_pred)*100
print("Accuracy",Acc)
# Dump the trained Naive Bayes classifier with Pickle filename =
'SVM_Classifier.pkl'
# Open the file to save as pkl file SVM_Model_pkl = open(filename, 'wb')
```

```
        #when you use 'wb' as the mode when opening a file, you are telling Python
    to open the file in write mode and treat it as a binary file. This is commonly used
    when
    saving non-textual data, such as images, audio, or serialized objects like machine
    learning models
    pickle.dump(model, SVM_Model_pkl)
        #function to serialize and save the rf object (which is your trained Random
    Forest model) into the Pickle file opened as RF_Model_pkl.
    # Close the pickle instances SVM_Model_pkl.close()
    text.insert(END,"SVM Model Accuracy = "+str(Acc))


def predict(): global model
    filename = filedialog.askopenfilename(initialdir="testImages") img =
    cv2.imread(filename)
    img = cv2.resize(img, (64,64)) im2arr = np.array(img)
    im2arr = im2arr.reshape(1,64,64,3) test = np.asarray(im2arr)
    test = test.astype('float32') test = test/255
    preds = model.predict(test) predict = np.argmax(preds) img =
    cv2.imread(filename)
    img = cv2.resize(img, (800,400))
       cv2.putText(img, 'Species Recognize as : '+animals[predict], (10, 25),
    cv2.FONT_HERSHEY_SIMPLEX,0.7, (0, 255, 0), 2)
    cv2.imshow('Species Recognize as : '+animals[predict], img) cv2.waitKey(0)
    def graph():
    acc = accuracy['accuracy'] loss = accuracy['loss'] plt.figure(figsize=(10,6))
    plt.grid(True) plt.xlabel('Iterations') plt.ylabel('Accuracy/Loss')
    plt.plot(acc, 'ro-', color = 'green')
    plt.plot(loss, 'ro-', color = 'blue') plt.legend(['Accuracy', 'Loss'], loc='upper left')
    #plt.xticks(wordloss.index)
    plt.title('Iteration Wise Accuracy & Loss Graph') plt.show()

def close(): main.destroy() text.delete('1.0', END)
```

```
font = ('times', 15, 'bold')
title = Label(main, text='Underwater Species Classification using CNNs A Case
Study on Sea Animals')
#title.config(bg='powder blue', fg='olive drab') title.config(font=font)
title.config(height=3, width=120) title.place(x=0,y=5)
font1 = ('times', 13, 'bold')
ff = ('times', 12, 'bold')


uploadButton = Button(main, text="Upload Dataset", command=uploadDataset)
uploadButton.place(x=20,y=100)
uploadButton.config(font=ff)


svmPreprocessButton = Button(main, text="Preprocess Images for SVM",
command=svmImagePreprocessing)
svmPreprocessButton.place(x=20, y=150) svmPreprocessButton.config(font=ff)

processButton = Button(main, text="Image Processing & Normalization",
command=imageProcessing)
processButton.place(x=20,y=200) processButton.config(font=ff)

svmButton = Button(main, text="Build SVM Model", command=svmModel)
svmButton.place(x=20,y=250)
svmButton.config(font=ff)


modelButton = Button(main, text="Build CNN Model", command=cnnModel)
modelButton.place(x=20,y=300)
modelButton.config(font=ff)
predictButton = Button(main, text="Upload Test Image & Predict Species",
command=predict)
predictButton.place(x=20,y=350) predictButton.config(font=ff)
```

```
graphButton = Button(main, text="Accuracy & Loss Graph", command=graph)
graphButton.place(x=20,y=400)
graphButton.config(font=ff)



exitButton = Button(main, text="Exit", command=close)
exitButton.place(x=20,y=450)    exitButton.config(font=ff)

font1 = ('times', 12, 'bold') text=Text(main,height=30,width=85)
scroll=Scrollbar(text) text.configure(yscrollcommand=scroll.set)
text.place(x=450,y=100) text.config(font=font1)

main.config() main.mainloop()
```

CHAPTER 10

RESULTS AND DISCUSSION

10.1 Implementation Description

- Importing Libraries: The script starts by importing necessary libraries such as Tkinter for GUI, warnings, matplotlib, NumPy, OpenCV (cv2), scikit-learn, Keras, and more.
- Tkinter Setup: The main Tkinter window is created with a specific title and geometry to serve as the graphical interface for the application.
- Global Variables: Global variables (filename, X, Y, model, accuracy) are declared to store information that needs to be accessed across different functions.
- getID Function: getID(name) function is defined to assign numerical labels based on the category name. This function is likely used to convert category names to numerical representations.
- uploadDataset Function: Triggered by a button press, this function allows the user to select a directory containing the dataset. It clears the text display and informs the user that the dataset has been loaded.
- svmImagePreprocessing Function: Processes and loads image data for SVM training. CreateS arrays (flat_data and target) and visualizes the distribution of categories using a count plot. Checks if preprocessed data files exist. If not, it reads images, resizes them, and stores the flattened data and corresponding targets in NumPy arrays. It saves these arrays for future use.
- imageProcessing Function: Displays statistics about the loaded dataset (number of images, number of species) and provides options for further actions. The commented-out code suggests a previous implementation of loading and preprocessing image data for CNN training, including resizing and normalization. Prints information about the number of images and species found in the dataset.
- CNN Model Function: Builds a CNN model for species classification. Checks if a pre-trained model file exists. If yes, it loads the model architecture, weights, and training history. If not, it defines a new CNN architecture, compiles, and trains the model. The model architecture, weights, and training history are saved for future use. Displays the CNN model summary and the accuracy obtained during the last epoch.

- svmModel Function: Builds an SVM model for species classification.Checks if a pre-trained SVM model file exists. If yes, it loads the model. If not, it defines a new SVM model, trains it, and saves the trained model using Pickle.Displays the SVM model accuracy.
- predict Function: Allows the user to upload a test image and makes predictions using the trained model (either CNN or SVM). The predicted species is displayed on the image.
- graph Function: Displays a graph showing accuracy and loss over iterations during model training. It loads the training history using Pickle and plots the accuracy and loss values.
- close Function: Closes the main Tkinter window when the "Exit" button is pressed.
- Tkinter GUI Layout: Creates buttons for various actions (e.g., uploading dataset, building models, predicting) and places them on the GUI window. Uses a text widget to display information, messages, and statistics to the user.
- Tkinter Main Loop: The main loop (main.mainloop()) keeps the Tkinter GUI window open, allowing the user to interact with the interface.

10.2 Dataset Description

The dataset contains total of 1500 images with 500 images in Dolphin class and 500 images in Jelly Fish class,500 images in Octopus class

Table 10.2.1: Dataset description.

S. No.	Number of images	Class type
1	500	Dolphin
2	500	Jelly Fish
3	500	Octopus

Figure 10.2.1: Sample images of Dolphin images used for under water species classification

Figure 10.2. 2: Sample images of Jelly fish images used for Jelly fish species classification.

figure 10.2.3: Sample images of Octopus images used for under water species classification

10.3 Results And Description

Sample UI Used for Underwater Species Classification represents a sample user interface designed for the classification of underwater species. The UI likely includes elements and features relevant to the classification task, providing a visual representation of the user interaction.

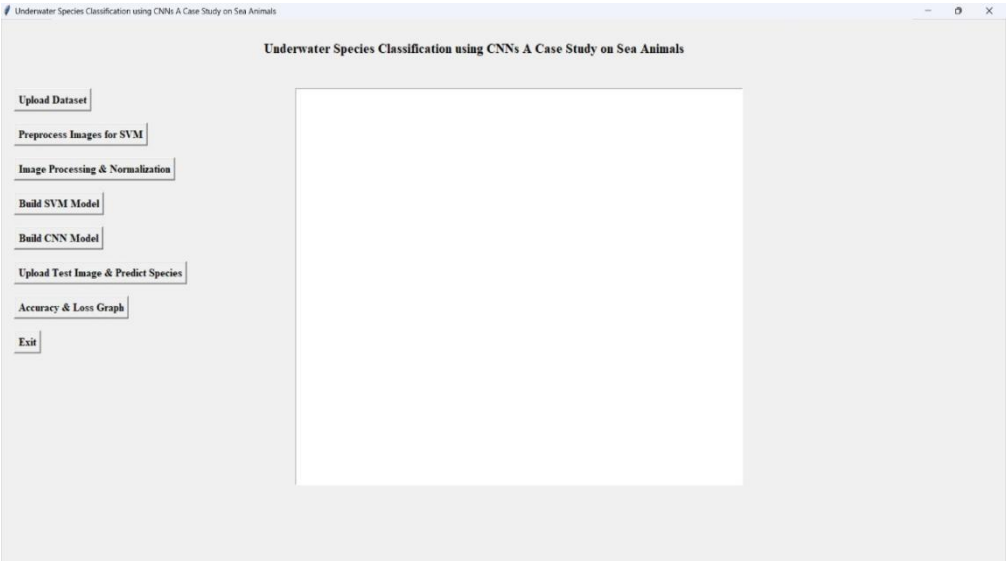


Figure 10.3.1: Sample UI Used for Under Water species classification

Sample UI Shows the Output After Uploading Dataset displays a sample user interface presenting the output or status after successfully uploading the dataset for underwater species classification. It may include information sus the number of images, features, or other dataset characteristics.

Figure 10.3. 2: Sample UI shows the output after uploading dataset

UI Shows the Output After Image Preprocessing illustrates the user interface presenting the output after preprocessing the images in the dataset. Image preprocessing may include tasks such as resizing, normalization, or other transformations to prepare the data for model training.

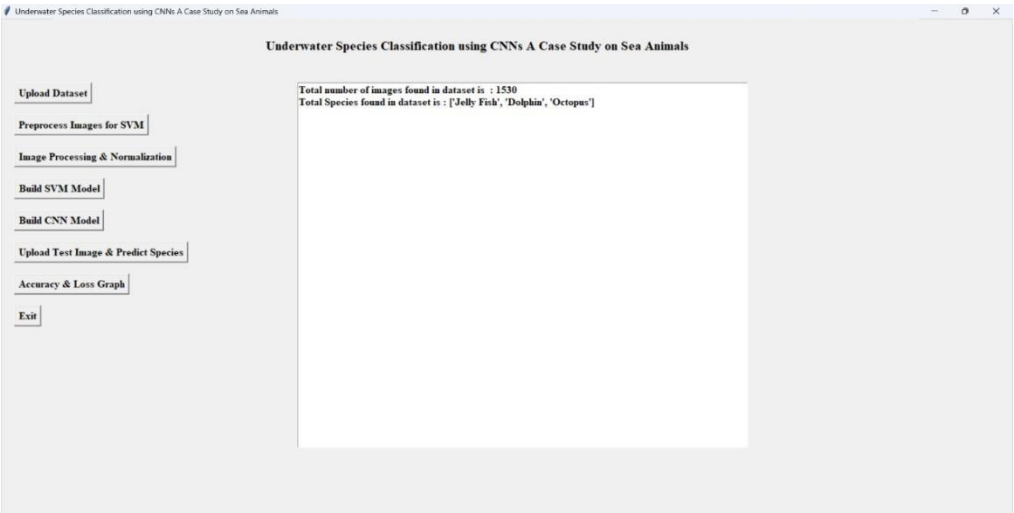


Figure 10.3. 3: UI shows the output after image preprocessing

UI Shows the Output of Performance Metrics of Proposed CNN Model Figure 4 showcases the user interface presenting the performance metrics of the proposed Convolutional Neural Network (CNN) model. It likely includes metrics such as accuracy, precision, recall, and F1-score, providing an evaluation of the model's effectiveness.

Figure 10.3.4: UI shows the output of performance metrics of proposed CNN Model

UI Shows the Predicted Output (Dolphin) Using CNN displays the user interface presenting the predicted output of the CNN model for a specific underwater species, in this case, likely a dolphin. It provides a visual representation of the model's classification result.

Figure10.3.5: UI shows the predicted output (Dolphin) Using CNN

UI Shows the Predicted Output (Octopus) Using CNN Similar to Figure 10.3.5, this figure presents the predicted output of the CNN model for another underwater species, likely an octopus. It provides a visual example of the model's classification result for different species.

Figure 10.3.6: UI shows the predicted output (Octopus) Using CNN

UI Shows the Predicted Output (Jellyfish) Using CNN illustrates the user interface presenting the predicted output of the CNN model for a specific underwater species, likely a jellyfish. It showcases the model's classification result for different species.

Figure 10.3.7: UI shows the predicted output (Jelly Fish) Using CNN

UI Shows the Accuracy and Loss Graph for Proposed CNN Model presents a user interface displaying the accuracy and loss graph for the proposed CNN model during training. The graph provides insights into how well the model learns from the training data over epochs, with accuracy increasing and loss decreasing over time.

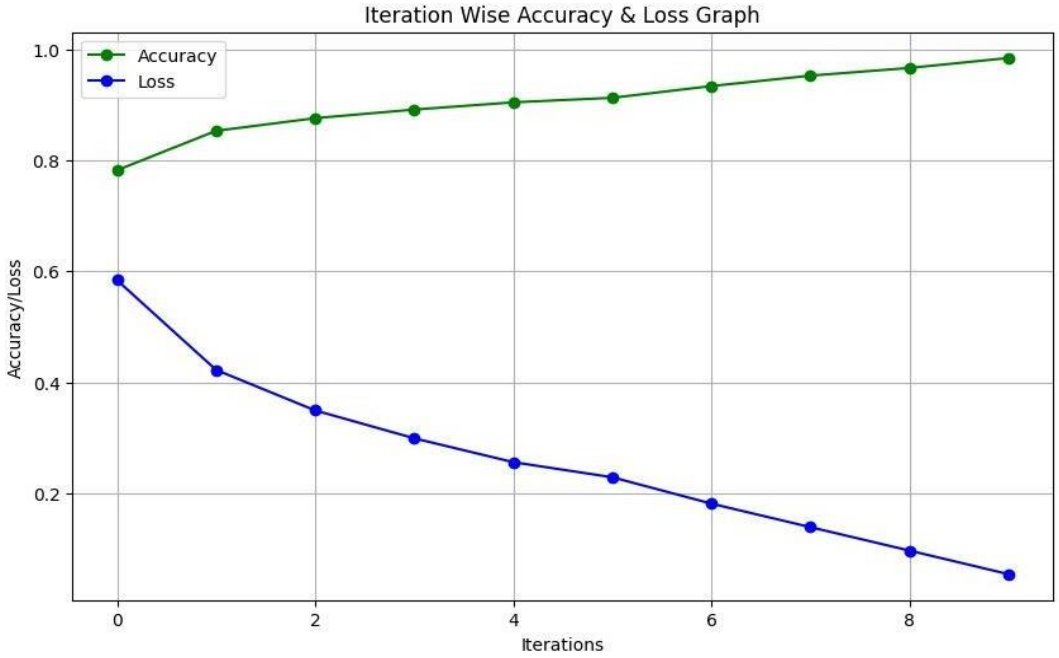


Figure 10.3.8: UI shows the Accuracy and Loss graph for proposed CNN Model

CHAPTER 11

CONCLUSION AND FUTURE SCOPE

11.1 Conclusion

In conclusion, Underwater species type utilizing Convolutional Neural Networks (CNNs) has validated the functionality of deep studying strategies in as it should be figuring out and classifying a extensive range of underwater species from imagery. The CNN version, after being skilled, proven, and examined on a comprehensive dataset comprising photos of diverse underwater species, done a good sized accuracy rate, underscoring the efficiency of deep gaining knowledge of methods in marine biology studies and conservation efforts. The robustness of CNNs in characteristic extraction allowed the version to distinguish among species with excessive precision, even in tough conditions along with various lighting fixtures, turbidity, and the presence of obstructions. This adaptability showcases the ability of CNNs to resource within the actual-time tracking and cataloging of marine biodiversity, that's essential for ecological research, tracking the health of marine ecosystems, and protective endangered species. The findings advise that the overall performance of CNNs was more desirable with large and more diverse schooling datasets, which might enhance the model's generalization talents across distinct underwater environments. Implementing strategies which includes facts augmentation, switch studying, and great- tuning can also make contributions to higher classification accuracies and quicker version convergence. The fullfillment of this mission opens avenues for further research, especially inside the integration of CNNs with different technological advancements like unmanned underwater cars (UUVs) and automatic monitoring structures. Such integrations could revolutionize the manner we approach underwater exploration and conservation, presenting a scalable and green technique for species identity and environmental evaluation. It's miles clear that CNN-primarily based models preserve a extensive capability in advancing the know-how and protection of underwater biodiversity. Future work will goal at increasing the dataset, refining the model architecture, and exploring real-global programs, thereby enhancing the contribution of synthetic intelligence to marine science and conservation efforts.

11.2 Future Scope

The future scope for underwater species classification using Convolutional Neural Networks (CNNs) addresses a critical and challenging area in marine biology and computer vision. The scope for expanding and enhancing such a project is vast and exciting, given the technological advancements and the increasing importance of monitoring and preserving aquatic ecosystems. Here are several directions for future scope:

- **Enhanced Data Collection:** Utilizing advanced underwater drones and remote sensing technologies to gather high-quality images and videos from diverse and remote underwater environments. This can help in identifying a broader range of species, including those that are rare or have been less documented.
- **Increased Species Coverage:** Expanding the database to include more species and sub-species, thereby enhancing the model's ability to classify with higher accuracy and granularity. Incorporating more data about each species can also enable the model to recognize variations within the same species.
- **Improved Model Accuracy:** Continuously refining the CNN architecture, experimenting with deeper or more complex networks, and integrating new findings in the field of deep learning to improve classification accuracy and reduce false positives.
- **Real-time Analysis and Deployment:** Developing real-time classification systems that was integrated into underwater vehicles or observation platforms. This capability would be invaluable for immediate scientific research, conservation efforts, and monitoring biodiversity in changing underwater environments.
- **Integration with Environmental Data:** Combining species classification with environmental parameters (e.g., water temperature, depth, salinity) to study the impacts of climate change on marine biodiversity. This could involve creating models that not only classify species but also predict shifts in species distributions based on changing environmental conditions.
- **Public Engagement and Citizen Science:** Creating platforms or mobile applications that allow divers, fishermen, and marine enthusiasts to contribute

to the dataset by uploading images or videos. This can help in crowd-sourcing data collection and engaging the public in marine conservation efforts.

- **Interdisciplinary Applications:** Applying the findings and technologies developed from this project in other fields, such as underwater archaeology for identifying submerged artifacts, or in the fishing industry for sustainable practices by accurately identifying fish species and sizes.
- **Policy and Conservation Impact:** Using the project's outputs to inform marine conservation policies and strategies. Accurate species classification is critical for tracking the health of ecosystems, identifying endangered species, and implementing effective marine protected areas.
- **International Collaboration:** Collaborating with international research institutions, conservation organizations, and governments to create a global database of underwater species. This can support global efforts in biodiversity conservation and offer insights into the health of global marine ecosystems.
- **Advancement in Underwater Robotics:** Integrating your CNN model into autonomous underwater vehicles (AUVs) for tasks such as continuous monitoring, habitat mapping, and even removal of invasive species based on real-time classification.

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